GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

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Abstract. We study the positive memory required to reproduce complete causal response laws. For every finite-group target, closed convex constraints on all command words admit a permutation realization at no greater memory cost. This reduces optimal uniform approximation over words of unbounded length to an attained finite optimization and gives robust multiplicity lower bounds. A sharp simultaneous-positivity inequality determines the complete feasibility region for bounded centered coefficient matrices with a common null vector. It yields a three-piece exact mean–contrast memory law for an unbounded symmetric-group family, together with boundary rigidity and optimal strategies. The observable group and predictive dimension stay fixed while the positive resource changes. For general finite-command quantum instruments, a physical return polytope and rare continuation branches give one finite simulator for every adaptive stopped experiment of bounded target expected cost. The memory depends on continuation-orbit growth; commuting branches give polylogarithmic bounds. For a fixed badly approximable qubit phase we prove a matching logarithmic law, with leading constant one, within the explicit class of physical-state dictionaries with Born terminal decoders. The inherited stationary initialization theorem separates the counted pure start from the continuing memory. Complete proofs of the general causal comparison, reset, optimal discrimination, variational, dual, spectral and geometric results are retained with their individual hypotheses.

1. Introduction

A predictive state contains the information in an experimental past that can affect future laws. Its linear dimension measures one kind of compression. A controller that stores probabilities on finitely many labels must also respect positivity, the preparation of its initial state, and the effect of subsequent observations. These constraints can make its minimum memory much larger than the predictive dimension. They also distinguish exact realization, uniform approximation of every command word, and approximation of experiments stopped at bounded expected cost. We study these distinctions within the original theory of causal resource comparison and optimal statistical discrimination.

A terminal response assigns a probability row $b(w)$ to each finite command word. An instrument response assigns joint probabilities to command–report histories and includes the state on which a terminal measurement acts. A classical realization uses one fixed collection of nonnegative transition and readout tables. Its initial label is pure and counted, except in explicitly named mixed-start comparisons. The physical quantum system, an external policy's storage, program bits, arithmetic precision and sampling work are separate resources. A private simulator label is never made visible to the external policy.

The central question is how an algebraic or geometric description of future responses controls their positive implementation. The present results give complementary answers. A finite-group specification admits a cost-preserving permutation reduction even when the required laws have

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uncertainty or error. A centered pair of nonnegative coefficient matrices has an explicit necessary and sufficient synthesis boundary, which gives the complete mean–contrast resource law of an unbounded nonabelian family. A return polytope and rare continuation branches give finite approximations of entire stopped experiments, with memory governed by the growth of the continuation semigroup.

Convex specifications and infinite-word approximation. Fix a homomorphism $\gamma : A^* \to G$ onto a finite group. For each $g \in G$ prescribe a closed convex set $K_g \subseteq \Delta(C)$ of allowed terminal laws. The requirement is imposed on *every* word with image g. Theorem 4.1 proves that an arbitrary stochastic controller on M labels satisfies these constraints if and only if there exist a finite G-set X, a preparation λ and a stochastic decoder D with

$$\lambda P_g D \in K_g \quad (g \in G), \qquad |X| + \mathbf{1}_{\{\lambda \text{ is mixed}\}} \le M.$$

For a free mixed start the last condition is simply $|X| \le M$. The original stochastic matrices are not assumed to satisfy group relations, and the resulting action need not be faithful. The proof averages identity words, projects onto closed communicating classes, and shows that the sandwiched action permutes their stationary rows. Closed convex constraints survive the limiting average. This extends the retained exact normal form by replacing an exact observable intertwining with a constraint-preserving projection.

Let $d_X(b)$ be the best uniform total-variation approximation to a target $b : G \to \Delta(C)$ on the action X, and let $d_X^{\mathrm{pt}}(b)$ restrict the preparation to a point. Theorem 4.2 gives the optimal error over all words and all M-label pure-start controllers:

$$\mathcal{E}_M^{\mathrm{p}}(b) = \min\left\{ \min_{|X|\le M} d_X^{\mathrm{pt}}(b), \min_{|X|\le M-1} d_X(b) \right\}. \tag{1.1}$$

Only finitely many actions occur. The minima are attained, depend Lipschitz-continuously on the target, and are computable by real quantifier elimination for algebraic data. This is synthesis for a supplied finite-group target, not an algorithm for unrestricted isolation or cutpoint questions about a supplied probabilistic automaton.

Theorem 4.3 supplies lower certificates. Fourier singular values larger than the allowed error force the corresponding irreducible multiplicities in the core. Below half the separation between distinct target residuals, any memory saving over their number must also pay for a mixed initializer. For a prime cyclic group, fewer than p labels do no better than the best constant law, whereas p labels are exact (Corollary 4.4). Approximation thus need not remove the algebraic memory obstruction when the error must be uniform over words of unbounded length.

A sharp boundary with two positivity constraints. The group average and the nonconstant response impose distinct constraints on a probability factorization. For $n \ge 5$ put $P = I - J/n$ and consider a probability column u and a real matrix A satisfying

$$0 \le A \le u\mathbf{1}^{\mathsf{T}}, \qquad \sum_{ij} A_{ij} = np, \qquad PAP = aP, \qquad \operatorname{rank}[A \mid u] < n.$$

The matrices A and $u\mathbf{1}^{\mathsf{T}} - A$ share a preparation constraint. Theorem 5.2 determines exactly which (p, a) permit such a pair, constructs a common nonnegative factor of width $n-1$ at every permitted pair of parameters, and classifies the coefficient pair on the curved boundary. This is an existence theorem; it does not claim that every given feasible pair has that factor width.

Put

$$p_0 = \frac{1}{n-\sqrt{n}}, \qquad p_1 = \frac{2}{n}, \qquad F_n(p) = \begin{cases} p(\sqrt{n}-1)/(\sqrt{n}+1), & 0 \le p \le p_0, \\ ((n-1)p-1)/((n-1)(np-1)), & p_0 \le p \le p_1, \\ (1-p)/(n-1), & p_1 \le p \le 1. \end{cases}$$

For $a > 0$, the answer is $a \le F_n(p)$. The proof combines the retained singular-matrix extremum with a simultaneous-positivity inequality. It permits signed null vectors and nonsymmetric coefficient matrices. The equality analysis forces the extremal coordinate pattern and then the allowable common factors. On the middle interval both outcome constraints are active; scaling the first-outcome bound alone would give an incorrect threshold.

Let χ_n be the natural permutation character of S_n, and set

$$\beta_{p,a}(g) = p + a(\chi_n(g) - 1), \qquad b_{p,a} = (\beta_{p,a}, 1 - \beta_{p,a}).$$

The exact probability range is $0 \le p - a \le 1$ and $0 \le p + (n-1)a \le 1$. For $a > 0$, Theorem 5.3 proves

$$\kappa_{S_n}(b_{p,a}) = \begin{cases} n(n-1), & a \le F_n(p), \\ n^2, & a > F_n(p), \end{cases} \qquad m(b_{p,a}) = 1 + \kappa_{S_n}(b_{p,a}).$$

Here κ is the least positive permutation core and m counts a pure start. Negative contrast exchanges the outcomes and replaces p by $1 - p$; zero contrast has one-label memory. At every nonzero contrast the observable group is S_n, the residual number is $n!$, and the predictive dimension is $1 + (n-1)^2$. The mean p is an average response over the group, not a prior on an unknown parameter. All minimum cores on the low and middle transition boundary are classified. The middle boundary forces equal orbit weights, while the low-mean boundary permits specified nonuniform weights. For $p \ge 2/n$, the lower resource suffices throughout the positive-contrast probability range. Two commands and binary readout realize the family as a prepared finite-dimensional quantum experiment.

The retained arbitrary-group results explain its representation-theoretic part: small-positive-contrast memory equals a permutation-module multiplicity cover (Theorem S3.2), and a single transitive response set is convex exactly at normal stabilizers (Theorem S3.3). These proofs, the exact R75 return law, the quotient linear programs and their payoff duals remain current inputs in the supplement.

Complete stopped laws and finite return geometry. Each observed instrument report changes the conditional state used by later calls. Theorem 7.2 assumes a finite physical polytope containing the preparation. Every returning branch maps the invariant state set into that polytope; continuing branches have total conditional probability at most $r < 1$. Retain all distinct normalized continuation images of the seed vertices of length less than L, and denote their number by $G(L)$.

One $G(L)$-label mixed-start model, or a counted pure-start model with at most $G(L) + 1$ labels, is exact before the first truncated excursion. Uniformly over visible adaptive command, stopping and terminal-POVM policies of target expected calls at most B, its complete stopped-law error is at most

$$\min\left\{1, \frac{(1-r)B}{r^{-L} - 1}\right\}. \tag{1.2}$$

The bound also holds for the terminal classical–quantum record, including nontermination mass. The exceptional-run estimate is sharp as a stopping-probability inequality (Lemma 7.1); this sharpness alone does not give a lower bound for every approximating controller.

With b commuting continuation maps and k seeds, $G(L) \le k\binom{L+b-1}{b}$. Other polynomial orbit-growth estimates give the corresponding power of $\log(B/\delta)$ for accuracy δ. The hypotheses concern individual normalized branches, and are verified directly in the application; average mixing is not substituted for a return condition.

For the retained undamped three-report qubit instrument, a regular triangle in the real Bloch plane contains every report-2 posterior; report 1 is a pure reset, and report 0 is a rotation of probability 1/8. For any fixed known alphabet of a commanded angles, Corollary 7.3 gives explicit

branch tables with

$$M \le 1 + 3\binom{L+a-1}{a}, \qquad L = \max\left\{1, \left\lceil \log_8\left(1 + \frac{7B}{8\delta}\right)\right\rceil\right\}.$$

One angle therefore needs at most $3L + 1$ labels, uniformly in its numerical value. This improves the damping upper for fixed finite alphabets. The earlier $O(\sqrt{B/\delta})$ construction still applies simultaneously to the full angle circle. These are different command quantifiers, not competing assertions about one unrestricted model.

Within the class of fixed physical-state dictionaries with Born terminal decoders, the one-angle upper has a matching lower. For a fixed badly approximable rotation phase and fixed $B > 0$, Theorem 7.4 proves

$$\lim_{\delta\downarrow 0} \frac{N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B)}{\log_8(1/\delta)} = \lim_{\delta\downarrow 0} \frac{N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B)}{\log_8(1/\delta)} = 1.$$

Rare reset–rotation words prepare separated pure states. Joint stopped record events force distinct physical dictionary points to approximate those states; zero-call readouts exclude the pure initializer from that packing. An orbit-prefix return polygon attains the leading constant. The golden-ratio phase has an explicit arithmetic certificate and five seed states. The original algebraic rotation has a rational-table upper with seven seeds, without an asserted badly approximable certificate. This lower is deliberately stated for physical dictionaries; arbitrary classical terminal decoders need not admit that representation.

The general stationary-initialization identity remains in the main article: except for the one-label case, a predictively stationary response costs exactly one label beyond its least mixed-start realization (Theorem 6.1). The complete noisy-instrument laws, exact sector and worst-phase counts, all-angle dictionary and boundary rigidity remain current in the supplement. Approximate truncation need not be stationary, so its extra initializer is an upper construction rather than an automatic lower bound.

Priority, dependencies, and the original causal programme. Residual equivalence and stochastic automata are classical [44, 79, 48, 49]. An exact deterministic residual theorem is not itself a new positive-realization principle. Rabin's isolated-cutpoint reduction and approximate matrix-product questions have different quantifiers from (1.1). Nonnegative stochastic groups [61, 62, 60, 66], finite-group representations [75], equivariant factorization and nonnegative-rank geometry [67, 68, 69, 70] provide established ingredients. Bounded simplex factorization and shared complementary-output factors are also prior theory [77]; generic nonnegative-rank thresholds and nested-polytope transitions have earlier examples [78]. Real-algebraic minimization of probabilistic automata is also prior work [65]; the finite-group projection makes the entire wordwise approximation constraint finite.

Polyhedral simulation and adaptive composition of quantum instruments have explicit antecedents [47, 76, 71]. Phase obstructions and noise-dependent classical-memory lower bounds are treated by Fanizza, Lumbreras and Winter [50]. Waiting-time martingales for patterns are classical [80]. The added claims are the stated convex-specification reduction, the simultaneous-positivity synthesis boundary, and the complete adaptive stopped-law construction from return geometry. A targeted literature search and finite diagnostics do not certify exhaustive priority.

The original actual-history comparison theorem and its common paid construction remain in the main article. Its witness is evaluated on the competitor's actual pre-call histories. The paired supplement keeps the complete arguments on performed-law compactness, variational and dual attainment, robust tied optimality, stopping rigidity, finite-width transfer, rank-constrained discrimination, finite-order quantum measurement, two-stage reset control, memory and communication, spectral geometry and optimal strategy implementation. The independent R58 companion retains calibration and nonintervening directional sensing. Local geometric conclusions keep their local hypotheses;

independent A1, A2 and further Theta objectives remain separate research problems. The theorem map distinguishes logical prerequisites of the current article from historical preservation.

2. Prepared kernels, acquired task laws and observers

All hidden-state, action and report spaces are standard Borel. A fixed unknown index $\theta \in \Theta$ specifies an actual preparation and positive normalized causal kernels. Positivity means nonnegative measures, not an everywhere strictly positive density. Dependence on the entire physical past may be included in the hidden state. A legal policy sees past visible reports and its admitted randomization, but not θ or a future audit. Iteration of these kernels defines its finite-prefix laws; the consistent countable path law is their kernel extension.

Each call has a cost $c_i \geq c_* > 0$. The decision to make the next call is predictable. In the geometric route, the declared productive-call flag is also predictable; in the primitive-instrument route, productive reports are specified outcomes and can be random. Commitment is permitted only after a productive report, and occurs exactly once almost surely in each model. At that time a specified future audit $Z \in H$ is scored, where $H \subset \mathbb{R}^k$ is nonempty, compact and convex. The prediction is committed before that audit is observed. A prospective joint report–audit kernel is defined at every productive call, including rejected calls. The stopping coin is independent of the audit conditional on the entire current visible report and the past. Write $L = \operatorname{diam}(H)^2$; the case $L = 0$ is trivial whenever a legal rule exists.

The core cost C_τ includes initial preparation charged to the core, failed calls and all productive calls through commitment. A separately executed final audit must be added to the raw budget. This convention is fixed before taking a risk infimum. Physical time, stored program constants and temporary computation are not identified with call count.

Three resource interfaces. A checkpoint procedure may use its whole precommit history, but must output one element of a fixed dictionary of at most M values. This counts its output width only; it is not a claim of $\log_2 M$ total working bits.

An autonomous observer has an alphabet S, $|S| \leq M$, and one fixed parameter-independent program. At an intercall boundary its only persistent observer memory is its label $s \in S$. Its next command distribution is a fixed function of s. A physical port name, elapsed time, subprogram index, random seed or former report is not an extra free input. During a call its current command, current report and fresh coins are available. Whatever persists to another call must be encoded in S. A fixed decoder $r : S \to H$ reads only the committed label.

In the *event-stopped* interface a post-report kernel emits $(s', e) \in S \times \{0, 1\}$. An event $e = 1$ ends the experiment and retains s' for the decoder; no subsequent command is executed. This public commitment event is not a stored bit used in continuing computation. The same label may be used live on one path and terminal on another. In the *state-stopped* interface a fixed map $h_* : S \to \{0, 1\}$ determines halt status. The initial label is live, and type-one labels are globally absorbing terminal labels with no executed command rows. Entry into a terminal label is allowed only after a productive report. Thus live and terminal types are disjoint in this interface, and only in this interface by definition.

Write $R_i^{\Theta}(N, M)$, $i \in \{\mathrm{cp}, \mathrm{ev}, \mathrm{st}\}$, for the infimum of worst-case expected squared loss over programs with robust almost-sure commitment and $\sup_\theta \mathbb{E}_\theta C_\tau \leq N$. An empty legal class has value $+\infty$. A program chosen once at random is not a free persistent seed. The ideal measurable class may have fixed real constants; effective implementation is stated separately in Section S40.

A physically acquired comparison law. Execute a declared prepared comparison experiment and obtain its joint report–audit probability $K_0(dx, dz)$ on $X \times H$. Its report marginal is ν. The law is genuinely preparable and its specification has its own resource account. It need not be a member

of Θ. In particular no finite set of calibration data is silently promoted to a uniform certificate. Set

$$f(x)=\int zK_0(dz\mid x),\quad w(x)=\int \|z-f(x)\|^2K_0(dz\mid x),\quad d_r(x)=w(x)+\|f(x)-r\|^2. \tag{2.1}$$

Disintegration gives Borel versions with $f\in H$; null sets may be completed arbitrarily. The profile is

$$\mathcal{Q}_m(b)=\min_{\substack{C\subset H,\ 1\le|C|\le m\\ 0\le h\le b,\ \int h\,d\nu=1}}\int [w+\operatorname{dist}(f,C)^2]h\,d\nu,\qquad b\ge 1,\ m\ge 1. \tag{2.2}$$

We use $+\infty$ for an infeasible cap or width. The quantities f, w are task coordinates of acquired predictive information, not a complete substitute for the predictive quotient. Their derivation and the quotient construction in Appendix S41 precede any use of local dimension.

Lower certificates on actual histories. Fix *one* actual model $\theta_*\in\Theta$, independent of the competing program. At each actual reached productive pre-call history h and legal command a, write $P_{h,a}(dy,dz)$ for its conditional full current report–audit law. Let $P^0_{h,a}$ be a measurable comparison row on the same report space, with a Borel mark $x=\pi_{h,a}(y)$ of law ν and

$$\mathbb{E}^0[Z\mid y]=f(\pi_{h,a}y),\qquad \mathbb{E}^0[\|Z\|^2\mid y]=w(\pi_{h,a}y)+\|f(\pi_{h,a}y)\|^2. \tag{2.3}$$

These are conditional identities given the *full* report, not just its mark. Versions and inequalities are required on actual reachable histories, not only almost everywhere under an unrelated comparison path law.

Fix $0<l_-\le 1\le u_-<\infty$. For all Borel $q: Y\to[0,1]$ and all finite-image Borel $r: Y\to H$, suppose predictable nonnegative defects $a_-(h,a), e_-(h,a)$ obey

$$\int q\,d(P_{h,a})_Y\le u_-\int q\,d(P^0_{h,a})_Y+a_-(h,a), \tag{2.4}$$

$$\int q(y)\|z-r(y)\|^2P_{h,a}(dy,dz)\ge l_-\int q(y)\|z-r(y)\|^2P^0_{h,a}(dy,dz)-e_-(h,a). \tag{2.5}$$

Independent current randomization can be adjoined to both rows: integrating inequalities for each fixed random value proves the same inequalities with the same defects. All command-conditioned lower inequalities hold before averaging the chosen command.

Let A_τ count productive calls. At this same actual model require, for every legal competing procedure,

$$\mathbb{E}_*C_\tau\ge\kappa_-\mathbb{E}_*A_\tau,\qquad \kappa_->0. \tag{2.6}$$

At budget N suppose, uniformly over all robustly feasible competitors,

$$\mathbb{E}_*\sum_{j\le\tau:\ j\ \text{productive}}a_-(H_{j-1},a_j)\le A_N,\qquad \mathbb{E}_*\sum_{j\le\tau:\ j\ \text{productive}}e_-(H_{j-1},a_j)\le E_N. \tag{2.7}$$

These are actual occupation certificates, not properties merely of the proposed upper program. Uniform row defects give $A_N\le(N/\kappa_-)\bar a_-$ and $E_N\le(N/\kappa_-)\bar e_-$. Lemma S38.6 supplies a nonuniform alternative.

One common upper construction. Independently, let $\mathcal{G}$ be a parameter-independent program which, from each of its start histories, reaches one productive report. Its conditional expected attempt cost is at most $\kappa_+>0$ on every surviving history in every $\theta\in\Theta$. It uses B live roles and returns to its start label on rejection; no reset of the physical state is assumed. It forms a virtual mark x, a current choice J and a permitted export T_Jr of its task readout. For every finite dictionary C assume

$$|\{T_jr: j\in J,\ r\in C\}|\le D_0|C|,\qquad D_0\in\mathbb{N}. \tag{2.8}$$

Every continuing phase and coin needed for this construction is included in the B live roles. For fixed $0 < l_+ \le 1 \le u_+ < \infty$ and $a_+, e_+ \ge 0$, assume for all Borel $q : X \to [0,1]$ and finite-image Borel $r : X \to H$, conditionally on every such history,

$$\mathbb{E}[q(x)] \ge l_+ \int q \, d\nu - a_+, \tag{2.9}$$

$$\mathbb{E}[q(x)\|Z - T_J r(x)\|^2] \le u_+ \int q(x) d_{r(x)}(x) \nu(dx) + e_+. \tag{2.10}$$

The expectation here includes the within-attempt choice J. Unlike the lower, the upper is not required to hold for each choice separately. The choices retained after this call are nevertheless charged through the actual export or live role. These hypotheses express one common causal construction, not modelwise optimizers.

3. Reversible response groups and common controlled memory

The fixed-phase results concern one transition matrix. Several commands require one family of matrices on the same retained labels, and checking every command in isolation is insufficient. We give an intrinsic finite-realization criterion for responses whose minimal linear dynamics is reversible and compact, including any finite family of unitary commands on a prepared finite-dimensional quantum system. We then obtain a matching arithmetic memory law for a pair of noncommuting involutions. This is a theorem about exact controlled response realization, not an assertion that a simulator's private label is an observed sufficient statistic for arbitrary feedback optimization.

Positive realization through invariant polyhedral sets is classical [48, 47, 46]. The argument below uses a section of the realizing simplex before passing to the observable quotient; auxiliary hidden-state responses need not belong to the minimal response space. Compact reversibility then turns containment into equality. The resulting finite-group obstruction concerns all command words, rather than a single spectral restriction.

3.1. **The minimal response space.**

Let A and C be nonempty finite alphabets of commands and terminal outcomes. A supplied response is a function $b : A^* \to \Delta(C)$. Its residual after a word w is $b^w(z) = b(wz)$, including the empty word $\varnothing$. Put

$$V_b = \operatorname{span}_{\mathbb{R}}\{b^w : w \in A^*\}, \qquad (L_a f)(z) = f(az).$$

Assume V_b is finite dimensional. It is invariant under each L_a, since $L_a b^w = b^{wa}$. Call the response *compactly reversible* if the restrictions $L_a|_{V_b}$ are invertible and the closure G_b of their generated group is a compact subgroup of $\mathrm{GL}(V_b)$. This concerns the observable quotient; it does not require the full physical command group to be finite or fully observable.

An exact stochastic realization with M live labels has row-stochastic matrices Q_a, one row-stochastic terminal decoder matrix D, and the prescribed initial row e_1^{T}, such that

$$b(a_1 \cdots a_n) = e_1^{\mathsf{T}} Q_{a_1} \cdots Q_{a_n} D. \tag{3.1}$$

Randomized terminal readout is allowed even when all Q_a are deterministic. The target input word is not encoded into an uncounted clock or persistent seed.

Theorem 3.1 (Finite realization of compactly reversible responses)**.** *For a compactly reversible response, the following are equivalent:*

(i) *An exact finite stochastic realization exists.*

(ii) *The residual set $\{b^w : w \in A^*\}$ is finite.*

(iii) *The observable group G_b is finite.*

(iv) *An exact realization with deterministic transitions and a possibly randomized decoder exists.*

The least number of deterministic-transition labels is the residual cardinality. This is not in general the least stochastic label count. If an M-label stochastic realization exists, then

$$|G_b| \le (2^M - 1)!.$$

This last bound is only a finite combinatorial certificate, not a sharp memory law.

Proof. Suppose (3.1) holds, and write $r_w = e_1^{\mathsf{T}} Q_w$. Let $R = \mathrm{span}_{\mathbb{R}}\{r_w\} \subseteq \mathbb{R}^M$. The assignment $\pi(r_w) = b^w$ extends to a well-defined surjective linear map $\pi : R \to V_b$: a relation $\sum_w c_w r_w = 0$ implies, after right multiplication by $Q_z D$, the identical relation among $b^w(z)$ for every z. Moreover

$$\pi(rQ_a) = L_a \pi(r) \qquad (r \in R).$$

The simplex section $K = R \cap \Delta_M$ is a compact polytope, invariant under right multiplication by every Q_a. Its image $P = \pi(K)$ is a compact polytope, contains every residual, and spans V_b. It satisfies $L_a P \subseteq P$. It is important that we intersect the simplex with R first: projecting arbitrary hidden-state responses directly into V_b would not be justified.

For an element T of a compact matrix group, positive powers have a subsequence tending to the identity with exponents tending to infinity. One elementary proof takes a convergent subsequence of powers with successively growing index gaps; quotients of two close powers approach the identity. A finite-order element is covered by its period multiples. Thus positive powers also approach T^{-1}. Applying this to L_a and using closedness of P gives $L_a^{-1} P \subseteq P$, and hence $L_a P = P$. All elements of G_b preserve P. They permute its finitely many vertices, and this permutation action is faithful because those vertices span V_b. Consequently G_b is finite. This proves (i)⇒(iii).

For the displayed bound, a vertex of K is determined by its positive support. If two distinct vertices had the same support, their difference could be added and subtracted in sufficiently small multiples while preserving R, unit mass and nonnegativity, contradicting extremality. Thus K has at most $2^M - 1$ vertices. A linear image of a polytope has no more vertices than the original polytope, so the faithful vertex action proves the bound.

If G_b is finite, the residual orbit is finite. Conversely, if the residual set is finite, each invertible L_a maps it injectively into itself, hence permutes it. Since it spans V_b, the generated group and its closure are finite. This proves (ii)⇔(iii). Under (ii), use the residuals as labels, send b^w to b^{wa} on input a, and use decoder $b^w(\varnothing)$. This realizes b with deterministic transitions. If any such deterministic-transition machine sends u and v to the same label, every suffix yields the same decoder law, so $b^u = b^v$. This proves its minimal count. Finally (iv)⇒(i) is immediate. No assertion that stochastic transitions can always be determinized without increasing size was used. □

Corollary 3.2 (Prepared unitary commands)**.** *Let U_a be finitely many unitary matrices on $\mathbb{C}^d$, let ρ be a prepared density matrix, and let $\{E_r : r \in C\}$ be a terminal positive-operator-valued measurement. For*

$$b(w, r) = \mathrm{tr}(E_r \Phi_w(\rho)), \qquad \Phi_a(X) = U_a X U_a^*,$$

exact finite stochastic realization for all command words is possible if and only if its observable response group is finite. The group is computed after quotienting only those directions invisible to every future commanded terminal test. Its dimension is at most d^2.

Proof. Let $W = \mathrm{span}_{\mathbb{R}}\{\Phi_w(\rho)\}$ in the Hermitian matrices. Each conjugation is a Hilbert–Schmidt isometry. Its forward invariant subspace W is also backward invariant, by finite-dimensional invertibility. Let

$$N = \{X \in W : \mathrm{tr}(E_r \Phi_z X) = 0 \text{ for all } z, r\}.$$

It is invariant under every Φ_a and hence under its inverse. The orthogonal complement of N in W is invariant as well. Therefore the induced transformations on W/N are orthogonal for the quotient inner product. The map $X \mapsto (z, r \mapsto \mathrm{tr}(E_r \Phi_z X))$ identifies W/N with V_b and intertwines them with L_a. Their generated group has compact closure in the orthogonal group. Apply Theorem 3.1. A constant measurement can make this induced group trivial even when the physical unitary group is infinite, so finiteness of the full physical group is not a necessary condition. □

3.2. **A matching memory law for two noncommuting commands.** Fix $x \in \mathbb{R}$ with $\sin(2\pi x) \neq 0$. On a protected qubit take

$$U_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_1 = \begin{pmatrix} \cos(\pi x) & \sin(\pi x) \\ \sin(\pi x) & -\cos(\pi x) \end{pmatrix}, \quad \rho = E = |0\rangle\langle 0|.$$

Both commands are involutions. For known $0 \leq \alpha \leq 1$, their channels are

$$\Phi_{a,\alpha}(X) = \alpha U_a X U_a^* + (1-\alpha)\operatorname{tr}(X) I/2.$$

Let $M_x^{\mathrm{ctrl}}(\alpha)$ be the least stochastic live memory of the whole binary response $w \mapsto \operatorname{tr}(E\Phi_{w,\alpha}\rho)$, using the same matrices and decoder for every word. In the real Bloch plane, with coordinates (z, x), these commands are αS_0 and αS_1, where S_0 reflects the angle t to $-t$ and S_1 reflects it to $2\pi x - t$. Their product is $\alpha^2 R_{2\pi x}$.

Theorem 3.3 (All-word arithmetic law for an involutive command pair)**.** *For $0 < \epsilon < 1$ and $\alpha = 1 - \epsilon$, the arithmetic envelope of Section S16 gives*

$$\tfrac{1}{4}\mathfrak{A}_x(\epsilon) \leq M_x^{\mathrm{ctrl}}(1-\epsilon) \leq (1+4\pi^2)\mathfrak{A}_x(\epsilon). \tag{3.2}$$

The upper is one explicit finite stochastic machine valid for every mixed command word, not separately chosen unary machines. The lower ranges over arbitrary common stochastic realizations.

For irrational x, the lower and upper logarithmic growth exponents as $\epsilon \downarrow 0$ are respectively

$$\frac{1}{\mu(x)+1}, \qquad \frac{\mu(x)-1}{2\mu(x)-1},$$

with values $0, 1/2$ at $\mu = \infty$. A uniform cubic-root law holds precisely for badly approximable x; otherwise the normalized count has lower limit zero and upper limit infinity.

At $\alpha = 1$, the exact count is the reduced denominator q if x is rational, and is infinite otherwise. At $\alpha = 0$, it is two. Every strictly damped pair has a finite exact common realization. These statements concern controlled response memory, not a general multi-command boundary-optimal policy theorem.

Proof. Any common realization restricted to the words $(01)^n$ realizes

$$\tfrac{1}{2}[1 + \alpha^{2n}\cos(2\pi n x)]$$

with transition matrix $Q_0 Q_1$. Its size is therefore at least $m_x(1-\alpha^2) \geq \mathfrak{A}_x(\epsilon(2-\epsilon))/2$ by Theorem S16.1. For $1 \leq c \leq 2$, the defining maximum of the envelope obeys $\mathfrak{A}_x(c\epsilon) \geq \mathfrak{A}_x(\epsilon)/c$ whenever both arguments are in its domain. This follows for each denominator before taking its infimum. Taking $c = 2 - \epsilon$ proves the lower in (3.2).

For the upper, the rational multiple-polygon construction in the proof of Theorem S16.1 gives a regular l-gon P on the unit circle, with vertex $v_0 = (1, 0)$, such that

$$l \leq (1+4\pi^2)\mathfrak{A}_x(\epsilon), \qquad \alpha R_{2\pi x} v_j \in P \quad \text{for every vertex } v_j.$$

The first reflection S_0 permutes this regular polygon and zero belongs to it. Consequently $\alpha S_0 P \subseteq P$, while $S_1 = R_{2\pi x} S_0$ gives $\alpha S_1 P \subseteq P$. Choose barycentric rows $Q_a(j,k)$ of $\alpha S_a v_j$ in the same vertices. Let $d_j = (1 + (v_j)_1)/2$. Starting at vertex zero, induction on the actual command word shows that the expected retained vertex equals the protected Bloch vector. Pairing with the first coordinate proves every response identity. The start is one of the counted vertices; there is no extra initialization mixture.

The two constants in (3.2) transfer both exponents and both uniform-law statements from Theorems S16.3 and S16.4. At zero damping, a rational phase of reduced denominator $q \geq 3$ has a common deterministic q-gon realization. The word-pair transition has a primitive qth-root mode and needs at least q states by the unit-circle spectral obstruction. For irrational x that same obstruction rules out every finite size. At $\alpha = 0$ the response is one at the empty word and $1/2$ after every

nonempty word, realized by a first label followed by an absorbing fair label; one label cannot change its decoder. This proves all endpoints. □

A fully algebraic example uses $U_1 = 5^{-1/2}\left(\begin{smallmatrix} 2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$. Then $e^{2\pi i x} = (3+4i)/5$ is not a root of unity, since its sum with its conjugate is the rational noninteger $6/5$. Each undamped command in isolation has period at most two; their common all-word response has no finite stochastic realization. The arithmetic exponent of this particular angle is not determined here. The example isolates the failure of commandwise testing without inferring a Diophantine class from finitely many coefficients.

If an independent herald with fixed probability p is placed before each command operation, and only failures apply the supplied command, every common realization above preserves the terminal response under every external command schedule and the first-herald cost $1/p$. The commands are actual inputs to that same retained-state transducer. This does not make its private label observable to the caller, allow uncharged midstream quantum measurements, or prove optimality for a controller free to choose an easier command sequence. The unary optimal squared-task interpretation below continues to use its separately proved physical embedding.

4. Convex specifications and optimal uniform approximation

The exact normal form has a robust extension. The target need not be an exactly specified probability table, and the stochastic controller need not satisfy the group relations, even on its observable states. What is required is a constraint on every word representing a given group element. The quantifier over all words is essential.

Fix a finite command alphabet A, a finite group G, and a surjective homomorphism $\gamma : A^* \to G$. For every $g \in G$ let K_g be a closed convex subset of the finite probability simplex $\Delta(C)$. Empty sets are allowed. An M-label pure-start controller consists of a prescribed point row e_o, stochastic command matrices Q_a, and a row-stochastic decoder D_0. It satisfies the specification if

$$e_o Q_w D_0 \in K_{\gamma(w)} \qquad (w \in A^*). \tag{4.1}$$

For a mixed start, replace e_o by an arbitrary probability row and do not charge an additional label for preparing that row.

Theorem 4.1 (Finite-group convex-specification normal form)**.** *The specification* (4.1) *has a pure-start realization on at most M labels if and only if there are a finite G-set X, a probability row λ on X, and a row-stochastic decoder D such that*

$$\lambda P_g D \in K_g \quad (g \in G), \qquad |X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \le M. \tag{4.2}$$

For mixed starts the corresponding criterion is $|X| \le M$. The action on X is permitted to have a kernel. In particular no faithfulness hypothesis is imposed on an approximating action.

Proof. We give the projection argument explicitly because its use of convexity replaces the exact observable intertwining in Theorem S1.1. Let $\mathcal{L} = \gamma^{-1}(e)$. Select finitely many identity words $\ell_1, \dots, \ell_s$ whose positive matrix entries cover every directed edge occurring in any Q_ℓ, $\ell \in \mathcal{L}$, and put

$$T = (I + Q_{\ell_1} + \cdots + Q_{\ell_s})/(s+1).$$

At most M^2 words suffice. Each closed communicating class C_i of T is closed under every identity-word matrix. The positive diagonal and Lemma S14.1 give a stochastic idempotent $E = \lim_n T^n$. Its image simplex has disjointly supported vertices $\mu_1, \dots, \mu_k$, the stationary rows of the closed classes. Every identity word preserves those classes, and consequently

$$EQ_\ell E = E \qquad (\ell \in \mathcal{L}). \tag{4.3}$$

For a word u write $F_u = EQ_uE$ on this image simplex. Choose an inverse word $\bar{u}$ with $\gamma(\bar{u}) = \gamma(u)^{-1}$. Since T^n is a convex combination of identity-word matrices, (4.3) applied to $u\ell\bar{u}$ and $\bar{u}\ell u$ gives

$$F_u F_{\bar{u}} = E = F_{\bar{u}} F_u.$$

Thus F_u is an affine automorphism of the image simplex and permutes its vertices. If $\gamma(u) = \gamma(v)$, multiplication by the common inverse $F_{\bar{u}}$ shows $F_u = F_v$. Replacing the middle E by T^n in EQ_uEQ_vE also gives $F_uF_v = F_{uv}$, since all words $u\ell v$ have the same group image. This defines a G-action P_g on $X = \{1, \dots, k\}$.

Write $e_oE = \sum_i \lambda_i\mu_i$ and put $D(i, \cdot) = \mu_i D_0$. For any representative u of g,

$$\lambda P_g D = e_oEQ_uED_0 = \lim_{n\to\infty} e_oT^nQ_uT^nD_0.$$

The last expression before passage to the limit is a convex combination of $e_oQ_{\ell u\ell'}D_0$ with $\ell, \ell' \in \mathcal{L}$. All these probability rows belong to K_g. Closedness and convexity therefore give $\lambda P_g D \in K_g$. This argument includes u empty and makes no claim that the original controller's word laws factor through G.

If the initial label lies in a closed class C_i, then $e_oE = \mu_i$ and λ is a point mass. Otherwise the k disjoint nonempty closed classes exclude at least the initial label, so $k + 1 \le M$. This proves the pure-start charge. For a mixed row η, use ηE instead and retain $k \le M$.

Conversely, a point preparation on X needs only the permutation transitions. For a mixed λ, a further label $\star$ has decoder λD and, on the first actual command a, moves into X with row $\lambda P_{\gamma(a)}$. Subsequent transitions are permutations. Its empty and nonempty word laws are respectively λD and $\lambda P_{\gamma(w)}D$. This gives the required realization and finishes both start conventions. □

For an exact probability table b, let $\kappa_G(b)$ be the least $|X|$ among factorizations $b(g) = \lambda P_g D$. This is its mixed-start minimum by the preceding theorem. The notation $m(b)$ retains the counted pure-start convention; its exact formula and the residual cardinality N are given in Theorem S1.2.

For a target $b : G \to \Delta(C)$ write $\mathrm{TV}(p, q) = \frac{1}{2}\sum_c |p_c - q_c|$ and define

$$\mathcal{E}^{\mathrm{p}}_M(b) = \inf_{\text{pure-start controllers on at most } M \text{ labels}} \sup_{w\in A^*} \mathrm{TV}(b(\gamma(w)), e_oQ_wD_0).$$

Use $\mathcal{E}^{\mathrm{m}}_M$ for the mixed-start convention. For a finite G-set X put

$$d_X(b) = \min_{\lambda, D} \max_{g\in G} \mathrm{TV}(b(g), \lambda P_g D), \qquad d^{\mathrm{pt}}_X(b) = \min_{x, D} \max_{g\in G} \mathrm{TV}(b(g), e_xP_gD).$$

The decoder ranges over stochastic rows; compactness makes both minima attained. Minima over no candidate actions are interpreted as $+\infty$.

Theorem 4.2 (Attained finite optimization for infinite-word error)**.** *For every target b, finite G, and integer $M \ge 1$,*

$$\mathcal{E}^{\mathrm{m}}_M(b) = \min_{|X|\le M} d_X(b), \tag{4.4}$$

$$\mathcal{E}^{\mathrm{p}}_M(b) = \min\left\{ \min_{|X|\le M} d^{\mathrm{pt}}_X(b), \min_{|X|\le M-1} d_X(b) \right\}. \tag{4.5}$$

One representative of each finite G-set is sufficient, so these are finite optimizations and their minima are attained. Their values are 1-Lipschitz functions of b in $\max_g \mathrm{TV}$. If $\mathcal{E}^{\mathrm{p}}_M(b) > 0$, a sufficiently small perturbation of b still cannot be exactly realized on M pure-start labels; the same assertion holds for mixed starts. When the target table is algebraic, the optimum and a realizing finite controller are computable by real quantifier elimination. This is a decidability assertion, not a polynomial-time bound or a claim that quantifier elimination was executed.

Proof. Apply Theorem 4.1 to the closed convex balls $K_g = \{p : \mathrm{TV}(p, b(g)) \le \delta\}$. It sends every controller of error at most δ to an admissible action without increasing the label charge or the error. Conversely every displayed action supplies such a controller. The pure constraint splits precisely into a point preparation with $|X| \le M$ or an arbitrary preparation with $|X| \le M - 1$. There are finitely many homomorphisms from the fixed finite group into each symmetric group of degree at most M. For each action the allowed preparations and decoders form compact sets. Hence the finite

minima are attained, including limits in which a mixed preparation becomes a point mass. The triangle inequality proves the Lipschitz bound and the perturbation assertion.

For the algorithmic statement, enumerate these actions and introduce variables $\lambda_x, D_{xc}, \delta$ and $u_{gc} \geq 0$ with

$$-u_{gc} \leq b(g,c) - (\lambda P_g D)_c \leq u_{gc}, \qquad \sum_c u_{gc} \leq 2\delta.$$

Together with the stochastic constraints these are polynomial inequalities over the real algebraic numbers. The point-start branch fixes $\lambda = e_x$ and enumerates x. The compact attained minimum is the least real point of the resulting semialgebraic set of feasible δ's. Real quantifier elimination and algebraic sample-point construction, used with the same scope as Theorem S8.2, compute the value and tables. □

The finite optimization concerns synthesis against a supplied finite-group target. It does not solve the cutpoint, equivalence, or isolation problem for a supplied arbitrary probabilistic automaton. Nor does it replace the specification for all words by a check of one representative per group element on the original stochastic controller. Those restrictions would change the problem.

Theorem 4.3 (Robust multiplicity and initialization certificates)**.** *Use the stacked Fourier matrix $A_\rho(b)$ of* (S3.1)*, with singular values in decreasing order, and write $m_\rho(X)$ for the multiplicity of ρ in $\mathbb{C}[X]$. If an action X approximates b to total-variation error at most δ, then*

$$\sigma_{m_\rho(X)+1}(A_\rho(b)) \leq \sqrt{2}\,\delta \qquad (\rho \in \widehat{G}). \tag{4.6}$$

A singular value past the number of columns means zero. For a binary response $b = (\beta, 1-\beta)$ the sharper equivalent scalar certificate is

$$\sigma_{m_\rho(X)+1}(\widehat{\beta}(\rho)) \leq \delta. \tag{4.7}$$

In particular let r_ρ^δ be the number of singular values of $A_\rho(b)$ strictly exceeding $\sqrt{2}\delta$. Then every mixed-start controller of uniform error at most δ uses at least

$$\min\left\{\sum_H q_H[G:H] : q_H \in \mathbb{Z}_{\geq 0}, \quad \sum_H q_H \geq 1, \quad \sum_H q_H m_\rho(H) \geq r_\rho^\delta \ (\rho \in \widehat{G})\right\} \tag{4.8}$$

labels. The scalar version may be used for binary outputs.

Let N be the number of distinct residual tables $b^h(g) = b(hg)$. If $N > 1$, let

$$\Delta_b = \min_{b^h \neq b^k} \max_g \mathrm{TV}(b(hg), b(kg)) > 0.$$

Whenever $\delta < \Delta_b/2$, a pure-start controller with fewer than N labels must have a projected mixed preparation. Its total budget is therefore at least one plus the lower bound (4.8)*.*

Proof. Put $a(g) = \lambda P_g D$ and $z_c(g) = b(g,c) - a(g,c)$. Since $\sum_c z_c(g) = 0$ and $\frac{1}{2}\sum_c |z_c(g)| \leq \delta$, the sums of the positive and negative parts are each at most δ, and $\sum_c |z_c(g)|^2 \leq 2\delta^2$. Unitarity gives

$$\left\| \begin{pmatrix} z_{c_1}(g)\rho(g^{-1}) \\ \vdots \\ z_{c_{|C|}}(g)\rho(g^{-1}) \end{pmatrix} \right\|_{\mathrm{op}} = \left(\sum_c |z_c(g)|^2\right)^{1/2} \leq \sqrt{2}\delta.$$

Average over g to obtain $\|A_\rho(b) - A_\rho(a)\|_{\mathrm{op}} \leq \sqrt{2}\delta$. The Fourier-multiplicity lemma gives $\operatorname{rank} A_\rho(a) \leq m_\rho(X)$. For completeness, if B has rank at most m, its nullspace intersects the span of the first $m+1$ right singular vectors of A in a nonzero vector. On a unit vector in that intersection, $\|(A-B)v\| = \|Av\| \geq \sigma_{m+1}(A)$. This proves (4.6); the binary version follows by averaging the scalar error, whose absolute value is at most δ, against unitary matrices. Summing orbit multiplicities and using the mixed-start normal form gives the integer bound.

For the last assertion, suppose the projected preparation were a point x. Its distinct residuals are indexed by points $x \cdot h$ in its orbit. If two target residuals b^h, b^k were represented by the same point, the two approximating residuals would coincide for every future g, and the triangle inequality would give distance at most 2δ between b^h and b^k. This contradicts the definition of Δ_b. The orbit must therefore contain at least N points. With fewer than N total labels this is impossible, so the projected start is mixed and its extra initializer is charged by the normal-form theorem. □

Corollary 4.4 (Uniform error cannot replace a prime orbit)**.** *Let $G = C_p$ with p prime and let $b : G \to \Delta(C)$ be any target. Put $r(b) = \min_{q \in \Delta(C)} \max_g \mathrm{TV}(b(g), q)$. For either start convention,*

$$\mathcal{E}_M(b) = \begin{cases} r(b), & 1 \le M < p, \\ 0, & M \ge p. \end{cases}$$

For binary outputs $r(b) = (\max_g \beta(g) - \min_g \beta(g))/2$.

Proof. Every orbit of C_p has size one or p. Thus any action with fewer than p points is trivial and gives a constant response, regardless of preparation. A one-label controller attains $r(b)$. On p points the regular action with a point preparation and decoder b is exact. For binary outputs the midpoint of the smallest interval containing all values of β is a Chebyshev center of that interval. □

For the symmetric-group target (S3.6), the scalar standard block is $tI/[n(n-1)]$ and $\Delta_{b_t} = 2t/n$. Indeed a permutation moving k points changes a residual by at most kt/n, with equality at an inverse representative; the smallest nonzero k is two. The subgroup-index estimate (S3.12) and the preceding theorem imply

$$\mathcal{E}_M^{\mathrm{p}}(b_t) \ge \frac{t}{n(n-1)} \qquad (n \ge 5,\ 0 < t \le 1,\ M \le n(n-1)). \tag{4.9}$$

If the error were smaller, the full standard multiplicity $n-1$ would be forced, and the small-error initialization charge would give at least $n(n-1)+1$ labels; here $n! > n(n-1)$. This explicit lower bound is not asserted to be the optimal error constant. At every exact resource jump the finite optimization also gives a strictly positive robustness radius, without a claimed closed formula for that radius.

5. The complete mean–contrast law and optimal-factor rigidity

The mean of a response and the size of its nonconstant part impose distinct positivity constraints. The return family of Theorem S3.6 fixes the mean at $1/n$. We now allow every feasible mean and both signs of contrast, for every $n \ge 5$. A second positivity inequality changes the sharp boundary: simply rescaling the old threshold gives the wrong answer on an interval of means. The proof yields the complete boundary factorization on that interval. Throughout, the mean is the average terminal probability over the group; it is not a prior on an unknown parameter.

Bounded simplex-structured factorization and stacking complementary nonnegative slacks with a common factor are established techniques [77, Section IV-C]. Threshold attainment and boundary identifiability also occur in the geometric theory of nonnegative rank [78, Section 4.3]. Our additional conclusion is the sharp simultaneous coefficient extremum below, its three-piece parameter region, and the resulting unrestricted group-memory and boundary-factor laws.

The following elementary extremum treats two nonnegative coefficient matrices sharing a left null vector. Both outcomes must be represented through the same preparation; this common-null-vector requirement is stronger than separate rank bounds on the two outcome matrices.

Lemma 5.1 (Sharp simultaneous positivity inequality)**.** *Let $n \ge 5$ and $v, c, d \in \mathbb{R}^n$ satisfy*

$$\sum_i v_i = 1, \qquad v \cdot c = v \cdot d = 0, \qquad c_i \ge v_j - \delta_{ij}, \quad d_i \ge \delta_{ij} - v_j \quad (1 \le i, j \le n).$$

Put $C = \sum_i c_i$, $D = \sum_i d_i$, and $\gamma_n = 2(n-2)/(n-4)$. For every real $\gamma \geq \gamma_n$,

$$\gamma C + D \;\geq\; L_n(\gamma) := \frac{\gamma(n+1) + n^2 - 2n - 1 + 2\sqrt{n(\gamma-1)(\gamma+n-2)}}{n-1}. \tag{5.1}$$

Equality holds exactly for the following triples, up to a simultaneous coordinate permutation. Set

$$m = \sqrt{\frac{\gamma+n-2}{n(\gamma-1)}}, \qquad M = \frac{1+m}{n-1}.$$

Then

$$\begin{aligned} v &= (M, \dots, M, -m), \\ c &= (M, \dots, M, M + M/m), \\ d &= (1-M, \dots, 1-M, (n-1)M(1-M)/m). \end{aligned} \tag{5.2}$$

In particular $1/\sqrt{n} < m \leq (n-2)/n$. No sign assumption on v or symmetry assumption on the resulting coefficient matrices is made.

Proof. The diagonal inequalities give $D \geq n-1$. If $v \geq 0$, the argument in Lemma S3.5 applies verbatim to c: a largest coordinate $M > 0$ forces all other positive coordinates to vanish, hence $C \geq n-1$. We verify below that $L_n(\gamma) < \gamma(n-1) - 1/(n-1)$ in the stated parameter range; therefore this case is strict.

Otherwise write $m = -\min_i v_i > 0$, relabel so that $v_1 = M$ is maximal, let L be the second largest coordinate, and put $b_0 = \max\{L, M-1\}$. The inequalities for c give

$$c = (b_0, M, \dots, M) + u, \quad u \geq 0, \qquad v \cdot (b_0, M, \dots, M) = M(b_0 + 1 - M) \geq 0.$$

Using $v \cdot u \geq -m \sum_i u_i$ and $v \cdot c = 0$ gives

$$C \geq b_0 + (n-1)M + \frac{M(b_0+1-M)}{m}. \tag{5.3}$$

This is the first-outcome estimate from the retained sharp singular-matrix proof. For the second outcome, $d = \mathbf{1} - v + z$ with $z \geq 0$ and $v \cdot z = \|v\|_2^2 - 1$. Consequently

$$D \geq n - 1 + \frac{1 - \|v\|_2^2}{m}.$$

The bound $-m \leq v_i \leq M$ implies $v_i^2 \leq (M-m)v_i + Mm$; summing it yields

$$D \geq n + \frac{1-M}{m} - nM. \tag{5.4}$$

This bound is valid even when its right side is below $n-1$.

As in the retained proof,

$$M \geq M_- := \frac{1+m}{n-1}, \qquad L \geq \frac{1-M+m}{n-2}, \qquad M_+ := 1 + \frac{m}{n-1}.$$

If $M \geq M_+$, (5.3) with $b_0 \geq M-1$ gives $C \geq nM - 1 \geq n-1$. Together with $D \geq n-1$ this case is again strict. If $M_- \leq M \leq M_+$, substitute $b_0 \geq (1-M+m)/(n-2)$ in (5.3); its right side increases with b_0. This gives

$$C \geq F(M) := \frac{1+m}{n-2} + (n-1)M + \frac{n-1}{m(n-2)}(M - M^2).$$

Therefore

$$\gamma C + D \geq G(M) := \gamma F(M) + n + \frac{1-M}{m} - nM.$$

The quadratic G is strictly concave in M, so its minimum on this interval is at an endpoint. At the left endpoint its value is

$$G(M_-) = \frac{\gamma(n+1) + n^2 - 2n - 1 + n(\gamma-1)m + (\gamma+n-2)/m}{n-1} \geq L_n(\gamma), \tag{5.5}$$

by the arithmetic–geometric mean inequality. At the right endpoint,

$$G(M_+) = (\gamma-1)nM_+ + n - \gamma - \frac{1}{n-1} \geq E_n(\gamma) := \gamma(n-1) - \frac{1}{n-1}.$$

For completeness, $E_n(\gamma) > L_n(\gamma)$ throughout the asserted range. After subtracting the rational terms and multiplying by $n-1$, this is equivalent to

$$n\{(n-3)\gamma - n + 2\} > 2\sqrt{n(\gamma-1)(\gamma+n-2)}.$$

The left side is positive. On squaring and dividing by n, the difference is

$$Q_n(\gamma) = n\{(n-3)\gamma - n + 2\}^2 - 4(\gamma-1)(\gamma+n-2).$$

For $n \geq 6$ its quadratic coefficient is positive, $Q_n(2) = n(n-6)(n-2) \geq 0$, and $Q_n'(2) = 2n(n-3)(n-4) - 4n - 4 > 0$. Hence it is strictly positive at $\gamma \geq \gamma_n > 2$. For $n = 5$, $\gamma_n = 6$ and $Q_5(\gamma) = 16\gamma^2 - 68\gamma + 57$ is positive and increasing for $\gamma \geq 6$. This proves the strict endpoint comparison and (5.1).

Equality forces $M = M_-$ and equality in (5.5); thus m has the stated value. The sum of the coordinates other than one minimum is $(n-1)M$, so each of them is M. Equality in (5.3) places every excess coordinate of c at the unique negative coordinate and gives $c_n = M + M/m$. Equality in the bound for D similarly places every excess over $\mathbf{1} - v$ at that coordinate, giving the stated d_n. The range $\gamma \geq \gamma_n$ is precisely $1/\sqrt{n} < m \leq (n-2)/n$, so $M + m \leq 1$. Thus $d_i = 1 - M \geq m$ for $i < n$, and $d_n \geq 1 + m$; all the required inequalities hold. The displayed c also satisfies its inequalities and $v \cdot c = 0$. Direct substitution verifies $v \cdot d = 0$ and equality in the bound. □

Let $\chi_n(g)$ count fixed points in the natural action of S_n. For a mean $p \in [0,1]$ and real contrast a, define

$$\beta_{p,a}(g) = p + a\{\chi_n(g) - 1\}, \qquad b_{p,a} = (\beta_{p,a}, 1 - \beta_{p,a}). \tag{5.6}$$

This is a probability table if and only if

$$0 \leq p - a \leq 1, \qquad 0 \leq p + (n-1)a \leq 1. \tag{5.7}$$

Both endpoint values occur, at a derangement and at the identity. Put

$$p_0 = \frac{1}{n - \sqrt{n}}, \qquad p_1 = \frac{2}{n}, \qquad F_n(p) = \begin{cases} p\,\dfrac{\sqrt{n}-1}{\sqrt{n}+1}, & 0 \leq p \leq p_0, \\ \dfrac{(n-1)p - 1}{(n-1)(np-1)}, & p_0 \leq p \leq p_1, \\ \dfrac{1-p}{n-1}, & p_1 \leq p \leq 1. \end{cases} \tag{5.8}$$

The three formulas agree at the two meeting points.

Theorem 5.2 (Sharp synthesis for bounded centered coefficients)**.** *Let $n \geq 5$, $P = I - J/n$, $0 < p < 1$ and $a > 0$. The following are equivalent:*

(1) *There are a probability column u and a real matrix A such that*

$$0 \leq A \leq u\mathbf{1}^{\mathsf{T}}, \qquad \sum_{i,j} A_{ij} = np, \qquad PAP = aP, \qquad \operatorname{rank}[A \;\; u] < n.$$

(2) *There exists such a pair admitting a common nonnegative factorization*

$$A = \sum_{i=1}^{n-1} w_i \lambda_i^{\mathsf{T}} D_i, \qquad u = \sum_{i=1}^{n-1} w_i \lambda_i^{\mathsf{T}}, \qquad w \in \Delta_{n-1}, \quad \lambda_i \in \Delta_n, \quad D_i \in [0,1]^n.$$

(3) $a \le F_n(p)$.

Thus joined ordinary rank deficiency and a positive factor of width $n-1$ have the same parameter-feasibility region in this centered coefficient problem. This does not assert that every feasible pair away from the boundary has that nonnegative factorization.

For $0 < p \le p_0$ and $a = F_n(p)$, all width-$n-1$ factorizations in (ii), up to factor and simultaneous coordinate relabeling, are

$$\lambda_i = \ell_i, \qquad D_i = \frac{np}{(n-1)w_i}\ell_i, \qquad \sum_{i=1}^{n-1} w_i = 1, \qquad w_i \ge \frac{np(1-1/\sqrt{n})}{n-1}, \tag{5.9}$$

where $\ell_i = (1-1/\sqrt{n})e_i + e_n/\sqrt{n}$ and D_i is the first-output decoder. For $p_0 \le p \le p_1$ and $a = F_n(p)$, all width-$n-1$ factorizations in (ii) have exactly the form

$$w_i = \frac{1}{n-1}, \qquad \lambda_i = (1-b)e_i + be_n, \qquad D_i = e_i + (np-1)e_n, \qquad b = \frac{p}{np-1}, \quad 1 \le i < n. \tag{5.10}$$

At $p = p_0$ these classifications agree. In the middle interval the coefficient pair (A, u) itself is unique up to simultaneous coordinate permutation; the proof gives it explicitly.

Proof. Start with any matrix pair in (i); no positive factorization is assumed in the converse. Put $B = u\mathbf{1}^{\mathsf{T}} - A \ge 0$. A nonzero common left null vector v of A and u also annihilates $B = u\mathbf{1}^{\mathsf{T}} - A$. If its coordinate sum were zero, $v^{\mathsf{T}}AP = av^{\mathsf{T}}$ would contradict $v^{\mathsf{T}}A = 0$. Normalize that sum to one. Exactly as in the retained null-vector decomposition,

$$A = a\{I + c\mathbf{1}^{\mathsf{T}} - \mathbf{1}v^{\mathsf{T}}\}, \quad B = a\{-I + d\mathbf{1}^{\mathsf{T}} + \mathbf{1}v^{\mathsf{T}}\}, \quad v \cdot c = v \cdot d = 0.$$

Nonnegativity gives all the inequalities of Lemma 5.1, and

$$C = \frac{p}{a}, \qquad D = \frac{1-p}{a}. \tag{5.11}$$

For $p \le p_0$, apply Lemma S3.5 to $A/(np)$: its centered coefficient is $a/(np)$, so its parameter there is a/p, giving $a \le p\tau_n = F_n(p)$. For $p \ge p_1$, the diagonal inequalities for d give $D \ge n-1$, hence $a \le (1-p)/(n-1) = F_n(p)$.

For $p_0 < p \le p_1$, set

$$m_0 = (n-1)p - 1, \qquad \gamma = \frac{nm_0^2 + n - 2}{nm_0^2 - 1}.$$

Then $1/\sqrt{n} < m_0 \le (n-2)/n$, so $\gamma \ge \gamma_n$, and m_0 is the minimizing value in (5.5). The equality triple (5.2) with $M = p$ has sums

$$C_0 = p(n + 1/m_0), \qquad D_0 = (1-p)(n + 1/m_0).$$

Thus $L_n(\gamma) = (\gamma p + 1 - p)(n + 1/m_0)$. Substituting (5.11) into the lemma and cancelling the positive factor $\gamma p + 1 - p$ gives

$$a \le \frac{1}{n + 1/m_0} = \frac{(n-1)p - 1}{(n-1)(np-1)} = F_n(p).$$

This proves the necessary boundary everywhere.

We give matching constructions at that boundary. In each case, smaller $a \ge 0$ is obtained on the same core by mixing the displayed decoder with the constant decoder p, with mixing weight $a/F_n(p)$. For $0 < p \le p_0$, use $n-1$ factors with equal weights, preparation rows ℓ_i and decoders $np\ell_i$. The largest decoder entry is $np(1-1/\sqrt{n}) \le 1$, precisely because $p \le p_0$. The coefficient

matrix is npA_*, where A_* is the retained sharp singular extremizer. Its mean is p and its centered part is $p\tau_n P = F_n(p)P$, as required.

For $p_0 \le p \le p_1$, use (5.10). Here $0 < b < 1$ and $0 < np - 1 \le 1$. Put $r = np - 1$ and $a_* = (1-b)/(n-1) = F_n(p)$. The resulting coefficient matrix has

$$A_{ii} = a_* \ (i < n), \quad A_{ij} = 0 \ (i \ne j < n), \quad A_{in} = a_* r, \quad A_{ni} = \frac{b}{n-1}, \quad A_{nn} = br = p.$$

The identity $r(nb-1) = 1$ shows that all cross differences of $A - a_* I$ vanish. Thus $PAP = a_* P$ and the group mean is p; the response is β_{p,a_*}.

For $p_1 \le p < 1$, put $b = 2/n$, use the same preparations $(1-b)e_i + be_n$ and equal orbit weights, and let

$$d_0 = \frac{np-2}{n-2}, \qquad D_i(j) = d_0 + (1-d_0)\mathbf{1}_{\{j \in \{i,n\}\}}.$$

These are valid decoders. At $p = p_1$ they agree with the preceding construction; mixing those decoders with the constant one shows that the response is $p + [(1-p)/(n-1)](\chi_n - 1)$ for every $p \ge p_1$. Equivalently this follows by direct permutation summation. This proves attainment at $F_n(p)$. The same constructions and decoder mixing give (iii)⇒(ii), while the displayed factorization immediately gives (ii)⇒(i). The converse already proved (i)⇒(iii), so the matrix synthesis equivalence is complete.

For the first rigidity statement, the equality case of the retained singular-coefficient lemma forces $A = npA_*$. The positive diagonal leading block and its zero off-diagonal entries force precisely one nonnegative rank-one summand at each of its $n-1$ diagonal positions. The adjacent last-row and last-column ratios then give (5.9), exactly as in the retained proof. Its decoder bound is $w_i \ge np(1 - 1/\sqrt{n})/(n-1)$. At $p = p_0$ summing these lower bounds forces equal weights. Pair uniqueness at this endpoint also follows without presupposing a positive factorization. The singular lemma fixes $A = np_0 A_*$, whose normalized left null vector is $(p_0, \dots, p_0, -1/\sqrt{n})$. Thus $v \cdot u = 0$ and $\sum_i u_i = 1$ force $u_n = 1/\sqrt{n}$. The bounds $u_i \ge A_{ii} = 1/(n + \sqrt{n})$ for $i < n$ then sum to $1 - 1/\sqrt{n}$, forcing equality in every one of them.

If $p_0 < p \le p_1$ and $a = F_n(p)$, equality holds in the weighted lemma with the chosen γ. It forces $v = (p, \dots, p, -m_0)$ and the unique c, d in (5.2). The coefficient matrix is therefore exactly the one displayed in the middle construction, while $u = a(c+d)$ has $u_i = a$ for $i < n$ and $u_n = b$. Its leading block again forces one rank-one summand at each diagonal position, with both factors supported on $\{i, n\}$. The last-row ratios force $\lambda_i = (1-b)e_i + be_n$. The diagonal entries force $D_i(i) = a/\{w_i(1-b)\} = 1/\{(n-1)w_i\}$. The upper bound $D_i(i) \le 1$ implies $w_i \ge 1/(n-1)$; summing the weights makes every one equal to $1/(n-1)$. The last-column ratios then give $D_i(n) = np - 1$. This proves the complete middle-boundary classification. □

Corollary 5.3 (Exact memory throughout the mean–contrast plane)**.** *Let $n \ge 5$ and let (p, a) satisfy* (5.7)*. At $a = 0$, both mixed-core and pure-start minima are one. If $a \ne 0$, put $q = p$ when $a > 0$ and $q = 1 - p$ when $a < 0$. Then*

$$\kappa_{S_n}(b_{p,a}) = \begin{cases} n(n-1), & 0 < |a| \le F_n(q), \\ n^2, & |a| > F_n(q), \end{cases} \qquad m(b_{p,a}) = 1 + \kappa_{S_n}(b_{p,a}). \tag{5.12}$$

Every nonconstant table has observable group S_n, residual cardinality $n!$, and observable dimension $1 + (n-1)^2$. For $a > 0$ and $p \ge 2/n$, the smaller memory realizes every feasible contrast. Thus positivity causes no additional core cost anywhere in that part of the parameter plane.

At the nontrivial boundary $a = F_n(p)$ with $0 < p \le 2/n$, all minimum permutation-core factorizations are exactly those in (5.9) *and* (5.10)*. For negative contrast, exchange the two output labels. This classifies minimum permutation cores, without asserting uniqueness of every unreduced stochastic machine.*

Proof. It suffices to treat $a > 0$, because exchanging the two outputs replaces (p, a) by $(1-p, -a)$. The trivial block of the Fourier transform is $(p, 1-p)$, and the nontrivial standard block is

$aI_{n-1}/(n-1)$ in the first coordinate. The faithful-standard-module argument in Theorem S3.6 therefore gives the claimed observable group, residual count and dimension for every $a > 0$. That theorem's subgroup-index estimate depends only on this module. It implies the lower bound $n(n-1)$ and proves that every core of size less than n^2 compresses to exactly $n-1$ natural orbits: extra orbits have no standard component, so their nonnegative constant contribution is absorbed into the natural-orbit decoders. This absorption preserves both output probabilities and does not use the value $p = 1/n$.

On n natural orbits, put weight $1/n$ on each, prepare point i on orbit i, and use

$$D_i(j) = p - a + na\,\mathbf{1}_{\{i=j\}}.$$

These decoders lie in $[0,1]$ exactly under (5.7). Their response is (5.6). Thus the core is always at most n^2, and there are no possible intermediate values.

Suppose a realization on $n-1$ natural orbits exists. Its preparation rows λ_i, weights w_i and decoder rows D_i give

$$A = \sum_{i<n} w_i \lambda_i^{\mathsf{T}} D_i, \qquad u = \sum_{i<n} w_i \lambda_i^{\mathsf{T}}, \qquad B = u\mathbf{1}^{\mathsf{T}} - A \geq 0.$$

Here $A \geq 0$, u is a probability column, and both coefficient matrices have the same preparation factor. Their permutation sums give the two outcomes. The permutation-switch argument in the retained proof implies $PAP = aP$, and averaging gives $\sum_{ij} A_{ij} = np$. The preparation matrix has at most $n-1$ rows, so $\operatorname{rank}[A \;\; u] < n$. The matrix criterion in Theorem 5.2 is therefore necessary. Conversely its width-$n-1$ positive factorizations give exactly $n-1$ natural-orbit preparations and valid binary decoders. Their coefficient identities imply the required response for every permutation. Hence a core below n^2 exists exactly when $a \leq F_n(p)$, and then the module lower bound makes its size $n(n-1)$. The retained pure-start formula gives $m = \min\{n!, 1+\kappa\}$, which equals $1+\kappa$ because $n^2 + 1 < n!$ for $n \geq 5$. At $a = 0$ the constant decoder uses one state. Every minimum core on the nontrivial boundary consists of $n-1$ natural orbits, so the factor classification follows from the matrix theorem. □

Physical realization and scope. Use the prepared unitary experiment of Theorem S3.6, with $\rho = n^{-1}\sum_i |ii\rangle\langle ii|$ and $E = \sum_i |ii\rangle\langle ii|$. Replacing the terminal effect by

$$E_{p,a} = (p-a)I + naE$$

produces (5.6); its two eigenvalues are $p-a$ and $p+(n-1)a$, so the exact validity range is (5.7), including negative contrast. The same two command generators realize every word. The theorem solves the whole displayed two-parameter family, including its constant boundary and its minimum-core factorization on the nontrivial transition curve. It is not an unrestricted high-contrast formula for every finite group, an identification result for an unknown model, or a solution of the original arbitrary-prior decision-risk problem.

6. The exact cost of initializing a stationary response

The distinction between a freely chosen initial distribution and a prescribed initial label is not peculiar to a periodic example. There is an exact general law. Its proof is an elementary elimination in the space of future response functions, within the classical positive realization framework [48, 47, 76]. It does not require hidden states to lie in the minimal linear realization, or require their unobserved transition matrix to be irreducible.

Let A be a nonempty finite command alphabet, with a finite report set Y_a for each $a \in A$. A finite classical realization consists of nonnegative matrices $T_{a,y}$ such that

$$T_a := \sum_{y \in Y_a} T_{a,y}, \qquad T_a\mathbf{1} = \mathbf{1},$$

and an initial probability row μ. For a command–report word $w = (a_1, y_1)\cdots(a_n, y_n)$ write $T_w = T_{a_1,y_1}\cdots T_{a_n,y_n}$ and $p(w) = \mu T_w \mathbf{1}$. Optional terminal readouts have decoder columns $d_{b,z} \geq 0$, with $\sum_z d_{b,z} = \mathbf{1}$, and response $p(w; b, z) = \mu T_w d_{b,z}$. All following equalities include every specified terminal readout when these are part of the experiment.

Call the prepared response *predictively stationary under* ν if ν is a probability row on A with every $\nu(a) > 0$ and

$$\sum_{a\in A} \nu(a) \sum_{y\in Y_a} p((a, y)w) = p(w) \qquad \text{for every future word } w. \tag{6.1}$$

When terminal readouts are specified, additionally require

$$\sum_{a\in A} \nu(a) \sum_{y\in Y_a} p((a, y)w; b, z) = p(w; b, z) \qquad \text{for all } w, b, z, \tag{6.2}$$

including the empty word w. This is an equality of observable responses; it does not assume $\mu \sum_a \nu(a)T_a = \mu$ in a particular hidden realization. For one command and no terminal readout it is ordinary stationarity. A response has a one-label realization precisely when its report probabilities factor as $p(w) = \prod_j p_{a_j}(y_j)$, and every terminal readout has a fixed distribution independent of the preceding word. For one command this is the i.i.d. case.

If a stationary hidden row is desired, it can be chosen without adding labels. Put $\overline{T} = \sum_a \nu(a)T_a$. Every $\mu\overline{T}^j$ has the same future response by the displayed stationarity identities. A subsequential limit of the Cesàro averages $H^{-1}\sum_{j=0}^{H-1} \mu\overline{T}^j$ is stationary for $\overline{T}$ and retains that response. Thus the freely mixed minimum below agrees with the convention requiring a stationary hidden row for the averaged command, and with the usual stationary HMM convention when there is only one command.

Let $n_{\mathrm{mix}}(p)$ be the minimum number of labels with an arbitrary initial probability row, and let $n_{\mathrm{pure}}(p)$ be the minimum when one prescribed initial label is counted. The minima are $+\infty$ when there is no finite realization.

Theorem 6.1 (Stationary initialization law)**.** *Suppose* (6.1) *holds, together with* (6.2) *when terminal readouts are included. If the response has a one-label realization, both minima are one. Otherwise*

$$n_{\mathrm{pure}}(p) = n_{\mathrm{mix}}(p) + 1, \qquad (+\infty) + 1 := +\infty. \tag{6.3}$$

The lower holds for all real nonnegative transition tables, including reducible and nonminimal linear realizations. The upper processes the first actual command while entering the core; it neither discards a physical call nor inserts an uncounted preliminary randomization.

Proof. Take any finite pure-start realization, with initial label o, and let f_i denote the complete future response from hidden label i. Thus $f_i(w) = e_i^{\mathsf{T}} T_w \mathbf{1}$, with the analogous values for terminal readouts. Put $\overline{T} = \sum_a \nu(a)T_a$. Stationarity gives the functional identity

$$f_o = \sum_j \overline{T}_{oj} f_j. \tag{6.4}$$

If $\overline{T}_{oo} = 1$, nonnegativity and full support of ν imply $(T_a)_{oo} = 1$ for every a. All report transitions from o then return to o. Their probabilities, and the terminal readouts at o, constitute a one-label realization. In the case under consideration this is excluded. Hence

$$\beta_j = \frac{\overline{T}_{oj}}{1 - \overline{T}_{oo}} \quad (j \neq o)$$

is a probability row and $f_o = \sum_{j\neq o} \beta_j f_j$.

Delete label o. On the remaining labels define

$$\widehat{T}_{a,y}(i, j) = T_{a,y}(i, j) + T_{a,y}(i, o)\beta_j, \qquad i, j \neq o, \tag{6.5}$$

and retain their terminal decoders. These matrices are nonnegative and their sums over y are stochastic. The recursion $f_i((a,y)w) = \sum_j T_{a,y}(i,j) f_j(w)$, with the functional identity for f_o substituted on its right side, is exactly the recursion for (6.5). Induction on word length, starting with the empty word and each terminal decoder, proves that every retained f_i is unchanged. Initial row β therefore realizes f_o with one fewer label. This proves $n_{\mathrm{mix}} \le n_{\mathrm{pure}} - 1$ whenever a finite pure realization exists.

Conversely, given a k-label mixed-start realization $(\mu, T_{a,y})$, append one label o. Its report transition on command a enters the core with row $\mu T_{a,y}$ and never returns to o; the core retains its original transitions. A terminal readout at o has row $\mu d_{b,z}$. The first physical command is already included in this row, so induction gives the original law at every word length, including zero. This uses $k+1$ labels and proves the reverse inequality. A finite pure realization is itself a finite mixed one, so the two finiteness assertions are equivalent. The one-label exception is immediate. □

Corollary 6.2 (Stationary processes)**.** *For every stationary finite-alphabet process that is not i.i.d., the minimum pure-start edge-emitting hidden Markov realization has exactly one more label than the minimum freely mixed realization, with the same infinite-value convention. In particular, the q versus $q+1$ conclusion of Theorem S12.1 is an instance of a general stationary principle, while its phase-dependent q remains a separate spectral conclusion.*

Equality of these complete response functions also preserves the law under every common randomized policy for commands, stopping and terminal readout. For each finite stopped history, multiply the equal response by the same policy probabilities. Summation gives all finite stopping events, and decreasing continuation events give the nontermination probability. Consequently all nonnegative costs measurable from that record have equal expectations, including infinite expectations. These are process-realization label counts. If the external policy has its own persistent memory, that memory is counted separately in a closed-loop implementation; it is not supplied free by the theorem.

7. Approximation by truncating rare conditional excursions

Exact memory can be infinite although one finite simulator approximates every stopped experiment with a prescribed expected cost. The damping construction of Theorem S7.1 gives such an upper bound for the entire angle circle. For a fixed finite command alphabet we now obtain a smaller bound by preserving a long conditional excursion exactly and modifying only its overflow. The general mechanism depends on the growth of the continuation semigroup, and applies beyond the qubit family.

The ingredients have classical antecedents. Polyhedral positive realizations are part of the theory of Heller and of Monràs and Winter [48, 47, 76]. Martingale calculations for pattern waiting times are classical [80]. The use made here is a uniform approximation of complete adaptive stopped laws, with an explicit memory bound from a finite return polytope and the number of distinct continuation states. It is separate from the exact noisy realization problem of [50].

7.1. **A sharp cost bound for a long conditional excursion.**

Consider sequential calls, each producing a report declared either *continuing* or *returning*. Conditional on the entire past, the chosen command and any private randomization, suppose the probability of a continuing report is at most r, where $0 < r < 1$. Let η_L be the first call completing L consecutive continuing reports, with $L \ge 1$. A policy can stop initially or after any call; its number of calls is τ.

Lemma 7.1 (Sharp expected-cost excursion bound)**.** *For every such policy with $\mathbb{E}\tau \le B < \infty$,*

$$\mathbb{P}\{\eta_L \le \tau\} \le \min\left\{1, \frac{(1-r)B}{r^{-L} - 1}\right\}. \tag{7.1}$$

The coefficient is optimal under the stated assumptions. For independent continuing indicators of probability r, the supremum over randomized stopping policies of expected cost at most B is exactly the right side.

Proof. Before the first completed run, let $K_t \in \{0, \dots, L-1\}$ be the current run length. It increases by one on a continuing report and returns to zero otherwise. Absorb it at L after the first completed run, and put

$$u(k) = \frac{r^{-k} - 1}{1 - r} \quad (0 \le k \le L).$$

For $k < L$ and any conditional continuation probability $p \le r$,

$$pu(k+1) + (1-p)u(0) - u(k) \le ru(k+1) - u(k) = 1.$$

Stop this calculation at $\sigma_H = \min\{\tau, \eta_L, H\}$. Summing the conditional inequalities, including the indicator that the next call is actually made, gives

$$u(L)\mathbb{P}\{\eta_L \le \tau,\ \eta_L \le H\} \le \mathbb{E}u(K_{\sigma_H}) \le \mathbb{E}\sigma_H \le B.$$

Let $H \to \infty$. Increasing events give (7.1); no assumption about the expected duration of another process is used.

For sharpness, take independent Bernoulli indicators of probability r. The first completed run has finite expectation: disjoint blocks of length L succeed with probability r^L, so its expectation is at most L/r^L. In this case the displayed drift equals one. Stopping at $\eta_L \wedge H$ and then using bounded convergence for $u(K)$ and monotone convergence for the duration proves $\mathbb{E}\eta_L = u(L)$. With probability $\min\{1, B/u(L)\}$, selected before making any call, run this experiment until η_L; otherwise stop immediately. Its expected cost is at most B and its success probability equals the right side of (7.1). □

Sharpness in this lemma concerns the probability of the exceptional run. It will not imply that every simulator must incur the approximation error estimated by that probability.

7.2. **A general finite return polytope construction.** Let $\mathcal{H}$ be finite dimensional and let $\mathcal{C}$ be a closed convex set of density matrices invariant under every positive-probability normalized branch below. The full state space is allowed. For each command in a nonempty finite set A, let $\{\mathcal{I}_{a,y} : y \in Y_a\}$ be a quantum instrument, with finite Y_a. Select a subset $R_a \subseteq Y_a$ of continuing reports. Suppose that there are density matrices $\sigma_1, \dots, \sigma_k$, with $P = \operatorname{conv}\{\sigma_1, \dots, \sigma_k\}$, such that

$$\begin{aligned} &\rho_0 \in P, \qquad \sum_{y \in R_a} \operatorname{tr} \mathcal{I}_{a,y}(\rho) \le r < 1, \\ &\frac{\mathcal{I}_{a,y}(\rho)}{\operatorname{tr} \mathcal{I}_{a,y}(\rho)} \in P \quad (y \notin R_a,\ \operatorname{tr} \mathcal{I}_{a,y}(\rho) > 0) \end{aligned} \tag{7.2}$$

for every command a and every $\rho \in \mathcal{C}$, with the seeds in $\mathcal{C}$. On the full state space, the continuation bound is equivalently the finite family of operator inequalities $\sum_{y \in R_a} \mathcal{I}_{a,y}^*(I) \preceq rI$. The return condition is a specified geometric hypothesis; it is not inferred merely from mixing of the unobserved channel. A prescribed preparation in $\mathcal{C}$ but outside P can be included by adding it to the seed list.

Write $\mathcal{E} = \{(a, y) : a \in A,\ y \in R_a\}$ and $b = |\mathcal{E}|$. For a word w over $\mathcal{E}$, let $\mathcal{I}_w$ be its chronological composition. For $L \ge 1$ define the finite set

$$\mathcal{D}_L = \left\{ \frac{\mathcal{I}_w(\sigma_j)}{\operatorname{tr} \mathcal{I}_w(\sigma_j)} : 1 \le j \le k,\ |w| < L,\ \operatorname{tr} \mathcal{I}_w(\sigma_j) > 0 \right\}, \qquad G(L) = |\mathcal{D}_L|. \tag{7.3}$$

The empty word is included. Repeated states may be identified.

Theorem 7.2 (Finite-memory approximation from return geometry)**.** *Assume* (7.2) *with* $0 < r < 1$. *For each* $L \geq 1$, *there is a single* $G(L)$*-label mixed-start classical model, chosen independently of the policy and of* B, *such that*

$$\sup_{\pi:\, \mathbb{E}^{\pi}_{\mathrm{target}}\tau \leq B} \|\mathcal{L}^{\pi}_{\mathrm{target}} - \mathcal{L}^{\pi}_{L}\|_{\mathrm{TV}} \leq \min\left\{1, \frac{(1-r)B}{r^{-L}-1}\right\}. \tag{7.4}$$

Here policies can choose commands, stopping and arbitrary terminal POVMs from the visible record and their own randomness. The complete stopped record includes all commands and reports, the stopping time, the terminal outcome and nontermination. The bound also holds in trace distance for the classical–quantum terminal record, with nontermination represented by its scalar mass. There is no additional entangled quantum register.

A counted pure start needs at most $G(L) + 1$ *labels. In general*

$$G(L) \leq k \sum_{j=0}^{L-1} b^{j}. \tag{7.5}$$

If the $b \geq 1$ *continuation maps commute pairwise, then*

$$G(L) \leq k \binom{L+b-1}{b}. \tag{7.6}$$

More generally any independently proved upper bound for the number of distinct normalized states in (7.3) *can be used in place of these bounds. If that number is* $O(L^{d})$, *the construction uses* $O((\log(B/\delta))^{d})$ *labels for fixed* r *and* $B/\delta \to \infty$.

Proof. Use the states of $\mathcal{D}_L$ as hidden labels. At a label ρ, the probability of report y on command a is $p_{a,y}(\rho) = \operatorname{tr} \mathcal{I}_{a,y}(\rho)$. Zero probabilities require no successor choice. For a returning report, choose any convex decomposition of its normalized output in the seed states. For a continuing report whose normalized output belongs to $\mathcal{D}_L$, use that output label deterministically. In the remaining case, send the state to any one fixed seed. Multiply each successor distribution by $p_{a,y}(\rho)$ to obtain the report transition row. All rows are nonnegative and the sums over the reports are stochastic. Use a seed decomposition of ρ_0 for the initial distribution. At label ρ, the terminal decoder for any POVM $(E_z)_z$ is $\operatorname{tr}(E_z\rho)$.

To verify the error, first use all finite continuation words, without the length restriction in (7.3). This countable refinement reproduces the target exactly. Every returning branch is expanded in the seeds; every continuing branch uses its exact normalized image. On each individual hidden label the weighted state identity is exact, so induction gives the target subnormalized quantum state for every command–report history. Initial mixing, terminal POVMs and common randomized policies preserve these identities. This is only a proof device, not an uncounted finite implementation resource.

Couple this exact refinement to the finite model by identical initial seed choices and by identical reports and successor choices while the states agree. The coupling first fails only when a continuing image leaves $\mathcal{D}_L$. After a return the exact label is a seed, so a product of fewer than L continuing branches is in $\mathcal{D}_L$ by definition. The same is true initially. Thus a failure requires a run of at least L consecutive continuing reports. Identifying duplicate states does not change this implication: their next normalized image depends only on the state itself, and every shorter word image is still present.

Before a failure the two visible histories, quantum states and policy choices can be identical. After a failure use arbitrary valid tails. For a fixed finite horizon, the stopped classical–quantum states have a common positive contribution from all paths with no failure. If the remaining mass is e, their trace distance is at most e, by positivity and the triangle inequality. The same assertion applies to the complete record after any terminal POVM. The target's probability of a failure before

stopping is at most $\mathbb{P}^{\pi}_{\mathrm{target}}\{\eta_L \le \tau\}$, which is bounded by Lemma 7.1. Its conditional continuation bound follows from (7.2), also after conditioning on the virtual seed choices.

For unbounded stopping, retain every stopped history up to horizon H and combine the unfinished histories into one symbol. The sum of trace norms on the finite stopped blocks increases to its full sum; the two unfinished masses decrease to their respective nontermination masses. Their absolute difference therefore converges to the nontermination term. The uniform finite-horizon bound passes to the limit, giving (7.4). The budget is always evaluated under the target. The argument does not require the approximate model to have expected duration at most B.

For a pure start append one initializer. On the first actual command it uses the initial mixture times the corresponding report matrix and enters the core. Its zero-call terminal decoder is that of ρ_0. It is never entered again. This proves the upper $G(L)+1$ without an uncounted preliminary randomization. If a separate finite policy uses s labels, the same construction gives at most $sG(L)+1$ labels for the closed product implementation with a prescribed initial label.

There are b^j words of length j, proving (7.5). For commuting maps the product depends only on a multi-index $m \in \mathbb{N}^b$ with $|m| < L$; there are $\binom{L+b-1}{b}$ such multi-indices. Normalization and deletion of zero images cannot increase this number. Finally it suffices to take

$$L \ge \max\left\{1, \left\lceil \frac{\log(1+(1-r)B/\delta)}{\log(1/r)} \right\rceil \right\} \quad (B>0,\ \delta>0).$$

The asserted consequence of polynomial orbit growth follows. □

The construction preserves every finite history containing no run of L continuing reports, including its unnormalized terminal state. It does not claim that its approximate response is stationary or that the extra pure-start label is necessary. When all continuation maps have finite normalized orbit on the seeds, the full finite orbit instead gives an exact model. These are upper constructions, not a converse classification of general positive realizations.

7.3. **A polylogarithmic simulator for finitely many commanded angles.** Apply the theorem to (S6.1) with $\alpha = 1$. Let the finite known angle alphabet be $\Theta = \{\theta_1, \dots, \theta_a\}$, where $a \ge 1$. The continuation report is 0, of probability 1/8. In the real Bloch plane put

$$v_0 = (1,0), \qquad v_1 = (-1/2, \sqrt{3}/2), \qquad v_2 = (-1/2, -\sqrt{3}/2).$$

Let σ_j be the corresponding pure states. Their triangle contains the disk of radius 1/2, and $I/2$ is their equal mixture. Report 1 returns to $\sigma_0 = \tau$. By Lemma S6.2, every normalized report-2 image has Bloch norm at most $5/11 < 1/2$, so it returns to this same triangle, at every angle and from every real-plane state. The real-plane density matrices form an invariant closed convex set: the unitaries are real, and the other two branches preserve real symmetric matrices. Thus they supply the set $\mathcal{C}$ in the general theorem.

Corollary 7.3 (Uniform finite-alphabet approximation)**.** *For the undamped instrument at any fixed finite angle alphabet Θ and initial state $I/2$, there is, for every integer $L \ge 1$, one classical model with at most*

$$3\binom{L+a-1}{a} \quad \textit{mixed-start labels, or} \quad 1+3\binom{L+a-1}{a} \quad \textit{counted pure-start labels,} \tag{7.7}$$

whose complete stopped-law error satisfies

$$\sup_{\pi:\, \mathbb{E}^{\pi}_1 \tau \le B} \|\mathcal{L}^{\pi}_1 - \mathcal{L}^{\pi}_L\|_{\mathrm{TV}} \le \min\left\{1, \frac{7B}{8(8^L-1)}\right\}. \tag{7.8}$$

The stronger stopped classical–quantum distance has the same bound. The statements include adaptive choices among all supplied angles and arbitrary terminal qubit POVMs.

For $B > 0$ and $\delta > 0$ it is enough to choose

$$L = \max\left\{1, \left\lceil \log_8\left(1 + \frac{7B}{8\delta}\right)\right\rceil\right\}. \tag{7.9}$$

For a single commanded angle this uses at most $3L+1$ pure-start labels, uniformly in that angle. For each fixed finite alphabet it uses $O_a((\log(B/\delta))^a)$ labels as $\delta/B \downarrow 0$. When the sine and cosine of each commanded angle are algebraic, every state and transition coefficient of the displayed construction is algebraic and is obtained by finitely many exact field operations.

Proof. The continuation maps are scalar multiples of commuting rotations, so their normalized orbit dictionary consists of

$$r_{j,m} = R_{m_1\theta_1+\cdots+m_a\theta_a} v_j, \qquad 0 \le j < 3, \quad m \in \mathbb{N}^a, \quad |m| < L. \tag{7.10}$$

We may keep all these indices, even if some represent the same state. Initialize uniformly on $(0,0),(1,0),(2,0)$. From a label with Bloch vector r, and command θ_i, set

$$h = \tfrac{1}{8} + \tfrac{1}{16} r_1, \qquad p_2 = \tfrac{7}{8} - h, \qquad u_2 = -\tfrac{1}{8} R_{\theta_i} r - h v_0.$$

Report 1 has a row of mass h into seed $(0,0)$. Report 2 has rows into the three seeds of masses

$$t_{2,j} = \frac{p_2 + 2 v_j \cdot u_2}{3}, \qquad j = 0,1,2. \tag{7.11}$$

Indeed, $\sum_j v_j = 0$ and $\sum_j v_j v_j^{\mathsf{T}} = \frac{3}{2} I_2$, so these masses sum to p_2 and their weighted Bloch vector is u_2. Moreover, $|u_2|/p_2 \le 5/11$ and $p_2 \ge 11/16$ give

$$t_{2,j} \ge \frac{p_2}{3}(1 - 10/11) \ge \frac{1}{48} > 0.$$

Thus the rows give the exact report-2 subnormalized state, with an explicit positive margin. On report 0, whose mass is $1/8$, move from (j,m) to $(j, m+e_i)$ if $|m| < L-1$. At $|m| = L-1$ send that mass to the three seeds uniformly. All report probabilities are unchanged on each represented input state, and their sum is one. Only the state following an overflow zero report is modified.

The resulting table uses words with the same multi-index identified at each depth and the same common-prefix proof as the theorem. It keeps the depth indices and may truncate at depth L even when a periodic phase makes that next physical state coincide with an earlier one. The first possible modified state still occurs on the Lth consecutive zero. Its state count is (7.7), and substitution of $r = 1/8$ in (7.4) gives (7.8). Formula (7.9) makes this bound at most δ. The label count and all error bounds are uniform in the numerical values of the finite supplied angles, although the transition table itself uses those values. The explicit rotations and (7.11) prove the algebraic assertion. □

If $B = 0$, a single initial decoder suffices, with no call and no error. The damping upper in Theorem S7.1 remains available and can be smaller for particular B, δ, a; one may choose the smaller of the two proved upper counts. Its simultaneous all-angle statement also remains valid. In contrast, the finite dictionary in (7.10) is for a *fixed finite* angle alphabet; the bound is not asserted for a continuum of possible commands.

For every non-root phase, exact finite classical realization is still impossible by Theorem S12.1. At that same fixed physical instrument the present bound is logarithmic in B/δ for a single angle. Thus an inverse-square-root exact noise law is not a lower bound on approximation of bounded-cost stopped experiments. Section 7.4 proves a matching logarithmic law within the explicitly restricted class of physical dictionaries; a matching lower for arbitrary classical terminal decoder tables remains open. The stopped-law bound also does not control the approximate model's expected duration: unbounded costs require a further tail or uniform-integrability hypothesis. External policy storage, table bits, arithmetic precision and exact sampling remain separate resources.

7.4. **A matching logarithmic law for physical dictionaries.** The preceding uppers use genuine density matrices as hidden dictionary states. Within that representation class a matching lower, including the leading logarithmic constant, is possible. This restriction is substantive: an arbitrary classical hidden-state model can have terminal decoder tables that are not Born probabilities of fixed density matrices. The conclusion below does not impose that restriction on such a model without proof.

Fix one commanded angle $\theta = 2\pi\omega$ in the undamped instrument. A *physical-dictionary simulator* has finitely many labels with fixed qubit density matrices σ_i, arbitrary stochastic report transition tables, and terminal decoder $d_E(i) = \operatorname{tr}(E\sigma_i)$ for every requested effect E. The causal transition tables are fixed independently of any later terminal-measurement request. The transitions themselves need not implement the quantum instrument on each dictionary state. Denote by $N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B)$ and $N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B)$ the least mixed- and counted-pure-start label counts that approximate, in total variation to error at most δ, every common visible stopping and terminal-POVM policy whose target expected calls are at most B.

Theorem 7.4 (Sharp leading order for a fixed badly approximable phase)**.** *Suppose ω is irrational and has a supplied constant $c > 0$ such that*

$$\|q\omega\|_{\mathbb{R}/\mathbb{Z}} \geq c/q \qquad (q \in \mathbb{Z}_{\geq 1}). \tag{7.12}$$

Necessarily $c \leq 1/2$. Fix $B > 0$ and put $\zeta = \min\{1, 7B/8\}$. For every integer $J \geq 2$, if

$$0 < \delta \leq \frac{\zeta}{8^J(1 + J^2/c^2)}, \tag{7.13}$$

then

$$N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B) \geq J, \qquad N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B) \geq J + 1. \tag{7.14}$$

There is also a finite integer K_ω, independent of B, δ, such that, with L from (7.9)*,*

$$N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B) \leq K_\omega + L - 1, \qquad N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B) \leq K_\omega + L. \tag{7.15}$$

Consequently, for either start convention and every fixed $B > 0$,

$$\lim_{\delta \downarrow 0} \frac{N_{\mathrm{phys}}(\delta; B)}{\log_8(1/\delta)} = 1. \tag{7.16}$$

For $\omega = (\sqrt{5} - 1)/2$, one may take $c = 1/3$ and $K_\omega = 5$. All approximation lower bounds in this theorem are for physical dictionaries with the stated Born terminal decoders.

Proof. For $n \geq 0$, consider the report word $1\,0^n$ and the pure target state ρ_n it prepares. Its Bloch vector is $v_n = (\cos(2\pi n\omega), \sin(2\pi n\omega), 0)$, with the coordinates ordered along the real Bloch plane. The word has probability $8^{-(n+1)}$. Use the following policy. With probability ζ begin the test; otherwise stop before any call. During the test stop as soon as a report differs from the specified prefix, or at its completion. Apply the terminal POVM $(\rho_n, I - \rho_n)$ on successful completion. The target expected number of calls is

$$\zeta \sum_{j=0}^{n} 8^{-j} \leq \zeta\, 8/7 \leq B.$$

Thus this is one of the policies in the definition of the approximation error, for every n, without a horizon-dependent loss of the budget.

In the simulator let $A_n = \sum_i a_{n,i}\sigma_i$, $a_{n,i} \geq 0$, be the subnormalized terminal state on the event that this test is begun and its complete prefix is observed. Put $p_n = \zeta 8^{-(n+1)}$. Two events in the full stopped record give

$$|\operatorname{tr} A_n - p_n| \leq \delta, \qquad \operatorname{tr}[(I - \rho_n)A_n] \leq \delta. \tag{7.17}$$

The second event has zero target probability. If $p_n > \delta$, these inequalities imply that some dictionary state with positive weight obeys

$$\operatorname{tr}(\rho_n \sigma_i) \geq 1 - \frac{\delta}{p_n - \delta}.$$

This conclusion is derived from joint events. No error bound for a rare conditional distribution has been assumed.

For $0 \leq n < J$, condition (7.13) gives

$$\frac{\delta}{p_n - \delta} \leq c^2/J^2.$$

If $r_i \in \mathbb{R}^3$, $|r_i| \leq 1$, is the Bloch vector of the corresponding physical dictionary state, then

$$|r_i - v_n|^2 \leq 2 - 2r_i \cdot v_n = 4[1 - \operatorname{tr}(\rho_n \sigma_i)] \leq 4c^2/J^2.$$

On the other hand, for $0 \leq m < n < J$, the elementary inequality $\sin(\pi x) \geq 2x$ on $[0, 1/2]$ and (7.12) give

$$|v_n - v_m| = 2|\sin(\pi(n-m)\omega)| \geq 4\|(n-m)\omega\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{4c}{n-m} > \frac{4c}{J}.$$

One dictionary vector cannot be within distance $2c/J$ of two such points. At least J distinct labels are therefore necessary.

For a pure start, the initial label has its own physical density matrix σ_o. Stopping before any call and choosing any binary terminal POVM is allowed at zero cost. The error condition therefore implies $\frac{1}{2}\|\sigma_o - I/2\|_1 \leq \delta$, or $|r_o| \leq 2\delta$. Its fidelity with any pure state is at most $1/2 + \delta$. Under (7.13), $J \geq 2$ and $c \leq 1/2$ give $\delta < 1/4$ and $c^2/J^2 \leq 1/16$. Thus the initial label cannot be one of the J labels whose fidelity is at least $15/16$. This proves both inequalities in (7.14).

For the improved upper, choose $K = K_\omega$ such that

$$P_K = \operatorname{conv}\{v_0, \ldots, v_{K-1}\} \supseteq \{x \in \mathbb{R}^2 : |x| \leq 5/11\}.$$

Such a finite K exists for every irrational ω. Here is a direct argument sufficient for this use. Pigeonholing finitely many fractional multiples of ω gives an integer $q > 0$ with $0 < d = \|q\omega\| < 1/4$. The points with indices $0, q, 2q, \ldots, \lfloor 1/d \rfloor q$ have successive circular gaps at most d, in one orientation or the other. Taking a prefix containing all these indices, its inscribed polygon has every central gap less than $\pi/2$, and hence inradius greater than $1/\sqrt{2} > 5/11$. The origin is in this polygon. It is a valid seed polytope: report 1 returns to v_0, and every report-2 output is in its contained disk.

For one angle, all continuation images of these K seeds through depth $L - 1$ belong to $\{v_0, \ldots, v_{K+L-2}\}$, a set of $K + L - 1$ states. Returning branches are expanded in the original K seeds. No overflow can occur before L successive zero reports, even when a return chooses the last seed. Theorem 7.2 therefore proves (7.15). The finite barycentric coefficients can be chosen from any triangulation of P_K. All labels, including the optional initializer representing $I/2$, are physical states.

For fixed B, the upper has $L = \log_8(1/\delta) + O_B(1)$. Conversely, (7.13) holds for an integer

$$J = \log_8(1/\delta) - 2\log_8\log_8(1/\delta) + O_{B,c}(1)$$

chosen sufficiently small, as follows by taking logarithms of that inequality. Dividing the two bounds by $\log_8(1/\delta)$ proves (7.16).

Finally put $\omega = (\sqrt{5}-1)/2$, and let p be a nearest integer to $q\omega$. Its conjugate is $\omega' = -(\sqrt{5}+1)/2$. The product $(q\omega - p)(q\omega' - p) = p^2 + pq - q^2$ is a nonzero integer. Therefore

$$|q\omega - p| \geq \frac{1}{|q\omega' - p|} \geq \frac{1}{\sqrt{5}q + 1/2} > \frac{1}{3q},$$

proving (7.12) with $c = 1/3$. The first five fractional orbit angles, in circular order, are

$$0, \quad 2\omega - 1, \quad 4\omega - 2, \quad \omega, \quad 3\omega - 1.$$

Their successive gaps are $2\omega - 1$ or $2 - 3\omega$. Since $3/5 < \omega < 5/8$, their maximum is $2\omega - 1 < 1/4$. This proves the required polygon inclusion with $K_\omega = 5$. □

The final limit fixes $B > 0$; it is not a uniform two-parameter asymptotic as B also varies. For arbitrary classical decoder tables, the general finite-alphabet upper remains valid, while the present packing lower need not apply. The badly approximable phase condition has an explicit quadratic-irrational instance above. It is not asserted for the separate algebraic rotation $(3 + 4i)/5$ studied earlier in the paper.

The improved upper does apply to that original algebraic rotation, with an explicit constant. Write $\theta = \arg((3 + 4i)/5) \in (0, \pi/2)$. The inequalities $\pi/6 < \theta < \pi/3$ put the first seven orbit points $0, \theta, \dots, 6\theta$ in their circular order with every gap less than π. The six consecutive edges have distance $\cos(\theta/2) = 2/\sqrt{5}$ from the origin. The closing edge has distance $-\cos(3\theta) = 117/125$. Both exceed $5/11$, so $K = 7$ is valid. This gives $L + 6$ mixed labels or $L + 7$ pure labels at the error in (7.8). All those orbit coordinates are rational. Triangulating their rational polygon gives rational initial and returning barycentric rows, so the resulting finite transition tables can be chosen rational. This is an explicit upper for that algebraic instrument, not a claim about its unresolved Diophantine lower certificate.

8. The causal task-order transfer theorem

Suppose that every terminating state-stopped competitor reaches at least $B_0 \geq 1$ live labels under θ_*. Such a certificate must concern actual physical command compatibility, not printed action names. Define

i	$m_-(M)$	$m_+(M)$
cp	M	$\lfloor M/D_0 \rfloor$
ev	M	$\lfloor M/D_0 \rfloor$
st	$M - B_0$	$\lfloor (M - B)/D_0 \rfloor$.

(8.1)

The event upper also requires $M \geq B$. The trivial state-stopped lower $B_0 = 1$ is valid for a procedure initially waiting to make a call.

Theorem 8.1 (Affine task-order transfer with a paid reference and common construction)**.** *Assume all the prepared-kernel, full-report, task-order, effort, occupation, resource and common-construction hypotheses of Section 2. Fix N, M, put $b_- = N/\kappa_-$, $b_+ = N/\kappa_+$, and suppose $b_- \geq 1$ and $m_-(M) \geq 1$. Set*

$$t_N = (1 - A_N)_+/u_-.$$

If $t_N > 0$, every one of the three risks satisfies

$$R_i^{\Theta}(N, M) \geq \left[l_- t_N \mathcal{Q}_{m_-(M)}(b_-/t_N) - E_N\right]_+. \tag{8.2}$$

If $t_N = 0$, the asserted lower is zero, with no evaluation of $\mathcal{Q}$ at infinity. An empty legal class still has risk $+\infty$.

Suppose also $m_+(M) \geq 1$, $b_+ \geq 1$, and, for event stopping, $M \geq B$. If

$$\beta_N = \frac{l_+ b_+}{1 + b_+ a_+} \geq 1, \tag{8.3}$$

one common Borel program with actual expected cost at most N in every model attains

$$R_i^{\Theta}(N, M) \leq \frac{u_+(1 + b_+ a_+)}{l_+} \mathcal{Q}_{m_+(M)}(\beta_N) + b_+ e_+. \tag{8.4}$$

It commits almost surely. If $\beta_N < 1$, the always-accept first-attempt construction instead gives $R_i^{\Theta}(N, M) \leq u_+ \mathcal{Q}_{m_+(M)}(1) + e_+$ at cost at most $\kappa_+ \leq N$.

The lower is against every legal competing strategy and uses one actual witness. The upper is one parameter-independent strategy for all models. The two effort constants need not agree. Neither

direction assumes that the comparison experiment belongs to Θ or that a normalized predictive update is uniformly contracting.

Corollary 8.2 (Scale-uniform relative transfer)**.** *Under Theorem 8.1, suppose $A_N = E_N = a_+ = e_+ = 0$. If $l_+ b_+ \geq 1$, then*

$$\frac{l_-}{u_-}\mathcal{Q}_{m_-(M)}(u_- b_-) \leq R_i^{\Theta}(N, M) \leq \frac{u_+}{l_+}\mathcal{Q}_{m_+(M)}(l_+ b_+). \tag{8.5}$$

The factors $l_\pm, u_\pm$ may be fixed nonunit constants, independent of budget and resolution. In particular no term proportional to $b_\pm$ times a fixed absolute calibration error is required in this relative-order class.

Proof. Substitute $t_N = 1/u_-$ and $\beta_N = l_+ b_+$ in the theorem. □

Corollary 8.3 (Separate exact checkpoint and autonomous laws)**.** *Suppose the certificates are exact: $l_\pm = u_\pm = 1$, all defects vanish, $\kappa_- = \kappa_+ = \kappa$, and exports are the identity. Then, for $N \geq \kappa$,*

$$R_{\mathrm{cp}}^{\Theta}(N, m) = \mathcal{Q}_m(N/\kappa).$$

The constructed autonomous upper uses $B + m$ state-stopped labels, or $\max(B, m)$ event-stopped labels. If the mandatory live count is sharp, $B_0 = B$, then

$$R_{\mathrm{st}}^{\Theta}(N, B + m) = \mathcal{Q}_m(N/\kappa) \quad (m \geq 1), \tag{8.6}$$

$$R_{\mathrm{ev}}^{\Theta}(N, M) = \mathcal{Q}_M(N/\kappa) \quad (M \geq B). \tag{8.7}$$

The subtraction by B belongs only to the state-stopped model. Nonidentity exports require the expanded dictionary in (8.1)*, unless an optimal dictionary is invariant under all exports. These exact stopping distinctions are inherited from the complete companion and are reproved here to fix the resource semantics.*

Proof. The same lower and upper profile coincide in each asserted interface. Identity exports give $D_0 = 1$. The packing and converse widths are proved in Lemma 9.4. No lower on live states is used to forbid event-stopped reuse. □

The familiar absolute-defect theorem is also a specialization: take $l_\pm = u_\pm = 1$ while retaining A_N, E_N, a_+, e_+. Thus relative and absolute calibration are composed in one statement. The chosen metric is the common scored task throughout; it is not changed between the converse and the construction.

9. Proof of the central theorem

Lemma 9.1 (Profile attainment)**.** *For every finite $m \geq 1$ and $b \geq 1$, the minimum in* (2.2) *is attained. The profile is nonincreasing in m, b and left-continuous in $b > 1$.*

Proof. Write a dictionary as an ordered tuple in the compact set H^m, permitting repetitions. The feasible densities are a weak-star closed subset of the radius-b ball of $L^\infty(\nu)$: positivity, the upper bound and the mass equality are each closed under pairing with $L^1(\nu)$. They are therefore weak-star compact. As the tuple varies, its squared-distance function varies uniformly on H, since all points belong to a bounded set. Along a convergent product net, split the objective difference into this uniform variation and the weak-star pairing against the fixed bounded measurable loss at the limit tuple. This proves continuity of the objective and attainment. Nets avoid an unwarranted sequential compactness assumption. Monotonicity follows from inclusion of feasible sets. For left continuity take an optimizer h at $b > 1$ and let $h_\lambda = (1-\lambda)h + \lambda$. It has unit mass and cap $b_\lambda = (1-\lambda)b + \lambda < b$. Its objective tends to the optimum as $\lambda \downarrow 0$, while monotonicity bounds the optimum at b_λ between the optimum at b and this objective. □

Lemma 9.2 (Actual-history comparison measure)**.** *Fix any robustly admissible competitor of cost at most N, with its fixed terminal dictionary C. There are finite nonnegative measures ξ_c on X, $c \in C$, such that $\xi = \sum_c \xi_c$ obeys*

$$\xi \le (\mathbb{E}_* A_\tau)\nu \le b_- \nu, \tag{9.1}$$

$$1 \le u_- \xi(X) + A_N, \tag{9.2}$$

$$\mathbb{E}_* \|Z - \widehat{r}\|^2 \ge l_- \sum_{c \in C} \int d_c \, d\xi_c - E_N. \tag{9.3}$$

The measure ξ is not asserted to be the terminal law of any comparison execution.

Proof. Adjoin an independent uniform random variable to the current report on both sides. This is current-call randomness, not a seed retained between calls; it is sufficient to represent the competitor's randomized output and stopping decision. A finite-output kernel has such a representation by cumulative probabilities. At a fixed reached productive history h and chosen command a, the post-report decision then gives an acceptance indicator $q_h(y)$ and an output $r_h(y) \in C$ on acceptance. Give r_h an arbitrary fixed value on rejection. The notation y now includes the current independent random variable. Integrating the original certificates over that variable preserves their constants. Dependence on the entire earlier history is permitted in these functions.

For each Borel $D \subset X$, set

$$\xi_c(D) = \mathbb{E}_* \sum_{j \le \tau:\ j \ mproductive} \int \mathbf{1}_{\{\pi_{H_{j-1},a_j}(y) \in D\}} q_{H_{j-1}}(y) \mathbf{1}_{\{r_{H_{j-1}}(y) = c\}} P^0_{H_{j-1},a_j}(dy, H). \tag{9.4}$$

The outer expectation is the actual pre-call occupation at θ_*. All terms are jointly measurable and nonnegative. Tonelli defines the measures, whose total is finite because $\mathbb{E}_* A_\tau \le b_-$. Summing over c and dropping the acceptance indicator gives (9.1): each comparison row has mark law ν.

The analogous sum with the actual rows instead of P^0 has mass exactly one, because the actual procedure commits exactly once almost surely. Apply (2.4) at each actual reached history, then sum. The occupation certificate yields (9.2). Apply (2.5) in the same way to the disjoint actual commitment events. Their expected losses sum to the actual risk. The comparison contribution is $d_{r_h(y)}(\pi_{h,a} y)$ by (2.3); this identity conditions on the full current report and therefore does not overlook extra information seen by the competitor. The accumulated loss defect is at most E_N, giving (9.3). One may first truncate the number of calls. Monotone convergence for the nonnegative contributions and the finite defect bounds justify the limit.

No actual past is replaced by a comparison past. In particular the competitor need not terminate, or obey a cost bound, if hypothetically run under the comparison experiment. This distinction is why the argument applies when that experiment is not an operating model. □

Remark 9.3 (Why the full report is required)*.* A constant mark can conceal a visible fair bit which also equals the future audit. A comparison using only that mark has $f = 0, w = 1$, but a two-value decoder reading the bit predicts it with zero loss. The hypotheses therefore condition on the full report seen by the competitor; matching a compressed mark marginal alone cannot establish the lower.

Lemma 9.4 (Live roles, output roles and physical compatibility)**.** *A program with ℓ live roles and t output roles has a state-stopped implementation with $\ell + t$ labels and an event-stopped implementation with $\max(\ell, t)$ labels, preserving every path cost and complete stopped scored law. Conversely an event-stopped M-label program has at most M decoder values. A state-stopped M-label program with at least B_0 reached live labels has at most $M - B_0$ terminal decoder values.*

For a physical live-state certificate, consider finitely many mandatory pre-call contexts I with no free context input. Suppose the commands legal at a history of context i are exactly $\mathcal{U}_i$, and every terminating competitor visits every context with positive probability under the same witness.

This positive probability may depend on the competitor; no uniform lower bound over competitors is required. Let b_{phys} be the least size of a cover of I by subsets J with $\bigcap_{i\in J}\mathcal{U}_i \neq \varnothing$. Then at least b_{phys} live labels are required. This certificate is invariant under duplicate names and bijective stateless recodings of the same physical commands.

Proof. For state stopping take disjoint copies of the live and terminal role sets and type them zero and one. For event stopping embed the live and terminal role sets separately into one alphabet of size $\max(\ell, t)$. On continuing transitions use the live embedding; on commitment emit the final event and use the terminal embedding. A command row is read only on a continuing branch, while the decoder is read only after commitment. Sharing their label therefore causes no conflict and uses no continuing mode bit. Couple the original and recoded coins. Every finite stopped prefix and every path cost agree, and almost-sure termination gives equality of the stopped scored laws. A fixed decoder has one value per retained label. In state stopping the global halt map excludes each reached live label from the output set, proving its subtraction. Event stopping has no such exclusion.

A live label reached in contexts J must have one command distribution supported on all the corresponding $\mathcal{U}_i$. Since J is finite, the union of the exceptional null command sets is null; hence their intersection supports a probability and is nonempty. The sets of contexts served by the labels cover I, proving the bound. Conversely a nonempty intersection contains a command and hence a Dirac command distribution. Replacing names without changing executed kernels, legal domains, costs and reports does not change these intersections. A translator which itself remembers a hidden phase is not such a recoding; its memory must be charged. □

Proof of Theorem 8.1. Fix a competing program. Lemma 9.4 bounds its decoder width by $m_-(M)$. Let $s = \xi(X)$ in Lemma 9.2. If $t_N > 0$, then (9.2) gives $s \geq t_N$. Scale the measure to $\xi' = (t_N/s)\xi$. It has mass t_N and remains bounded above by $b_-\nu$. Thus $h = (d\xi'/d\nu)/t_N$ has unit mass and cap b_-/t_N. Nearest-dictionary projection and nonnegativity imply

$$\sum_{c\in C}\int d_c\, d\xi_c \geq \int [w + \operatorname{dist}(f, C)^2]\, d\xi \geq t_N \mathcal{Q}_{m_-(M)}(b_-/t_N).$$

Together with (9.3) this proves the asserted lower for this competitor. Its worst-model risk dominates its risk at θ_*. Taking the infimum over competitors gives (8.2), without exchanging an infimum and a supremum. Nonnegative loss gives the zero truncation and the stated case $t_N = 0$.

For the upper assume (8.3). Choose an attained optimizer (C, h) at width $m_+(M)$ and cap β_N. Let $r_C(x)$ be the first nearest point to $f(x)$ in a fixed ordering of C. After each attempt of $\mathcal{G}$, accept x with probability $h(x)/\beta_N$, export $T_J r_C(x)$ and commit; otherwise return to the start routing label. All these operations are Borel. The choice J is used within the current attempt; any part retained thereafter is charged through the live role or exported decoder, not a free seed. The exports are assumed jointly measurable and take at most $D_0|C|$ distinct values, so their finite decoder and transitions are Borel as well.

Conditional on every surviving attempt history, the acceptance probability is at least

$$p = \frac{l_+}{\beta_N} - a_+ = \frac{1}{b_+} > 0. \tag{9.5}$$

Consequently the probability of reaching attempt $j+1$ is at most $(1-p)^j$, by induction on conditional survival probabilities. In particular commitment is almost sure and the expected number of attempts is at most $1/p = b_+$. Each begun attempt has conditional expected cost at most κ_+, not merely a cost conditional on a favorable outcome. Summing the predictable attempt costs gives actual expected cost at most $\kappa_+ b_+ = N$.

The conditional unnormalized loss contributed by an attempt is at most $u_+ \mathcal{Q}_{m_+(M)}(\beta_N)/\beta_N + e_+$. Sum its contributions on the disjoint terminal events, using the same survival bound. The result is

at most

$$b_+\left[\frac{u_+}{\beta_N}\mathcal{Q}_{m_+(M)}(\beta_N)+e_+\right],$$

which is (8.4). The program has B live roles and at most $D_0 m_+(M)$ terminal roles. Lemma 9.4 gives exactly the sufficient widths in (8.1). A checkpoint counts only the terminal width. No physical renewal law was used: the conditional hazard controls survival even if the hidden state after rejection differs arbitrarily from its former state.

If $\beta_N < 1$, commit after the first attempt, and use an optimizer at cap one. The loss certificate with $q = 1$ gives $u_+\mathcal{Q}_{m_+(M)}(1) + e_+$, and the whole-attempt cost is at most κ_+. This proves all remaining assertions. □

10. Scope and further mathematical questions

The finite-group convex-specification theorem is general over the stated finite alphabets, finite groups, closed convex constraints and arbitrary stochastic controllers. Its uniform approximation quantifier ranges over all finite command words. The resulting finite optimization is not a closed numerical formula for every group and target. The exact small-contrast multiplicity law and normal-stabilizer convexity criterion remain general within their stated classes. The complete mean–contrast formula concerns the displayed symmetric-group family for every $n \geq 5$; it does not classify high contrast for all finite groups. Its matrix synthesis theorem is an existence equivalence over parameter pairs, not a positive-factorization theorem for every given feasible matrix.

The return-polytope theorem applies to finite-command instruments on an explicit invariant physical state set, with a uniform conditional bound on continuing branches and a finite return polytope. These hypotheses are verified for the undamped qubit application. A fixed finite angle alphabet gives the new polylogarithmic upper; the retained damping construction works simultaneously for the entire angle circle. The matching logarithmic lower and leading constant require fixed $B > 0$, a fixed badly approximable phase, and physical dictionaries with Born terminal readouts. They do not apply to arbitrary classical decoder tables without a further argument. The original algebraic phase has an explicit rational-table logarithmic upper; its analogous Diophantine lower certificate is not supplied here.

The stationary initialization theorem applies to arbitrary predictively stationary finite-alphabet response laws and makes no quantum, reversibility, finite-rank or irreducibility assumption. The exact noisy instrument laws concern their stated family and spectral regimes. One-dependence and stationarity are assertions about a fixed command; adaptive command choices can transmit dependence. A private simulator label is not an observation available to the original policy. The approximating process need not satisfy the target's expected stopping budget: total variation alone does not control an unbounded expectation.

The general-M squared-sign minimax-risk frontier, unrestricted approximate-decoder lower bounds, arbitrary nonrigid robust feedback and tied boundary decisions, and the intermediate phase-type timing frontier remain distinct mathematical problems. None is replaced by a solved response-memory problem. Program length, arithmetic precision, working storage and exact sampling costs also remain distinct from live labels.

The actual-history comparison theorem and the independent causal, reset, discrimination and geometric results keep their original quantifiers. Local geometric conclusions remain local. Independent A1, A2 and further Theta construction objectives require their own proofs. The supplement contains the complete technical arguments and independent developments; all inherited mathematical bodies and the separate R58 companion have traceable preserved locations. Historical drafts and source-recovery records are available without becoming prerequisites for reading the current theorem chain.

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